Curriculum Tables

Department of Mechanical Engineering

Indian Institute of Technology Madras

Last updated: 17-November-2017

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Nomenclature

Students joined before 2015

L: Number of Lectures
T: Number of Tutorials
P: Number of Laboratory Sessions
C: Number of Credit

SCY: Basic Sciences-Chemistry
SPH: Basic Sciences-Physics
SMA: Basic Sciences-Mathematics
BET: Basic Engineering Theory
BES: Basic Engineering Practice
PMT: Professional Major Theory
HPF: Humanities Professional Compulsory
SLS: Life Sciences
PML: Professional Major Laboratory
HSS: Humanities and Scoial Sciences Elective
MNS: Minor Stream Elective
PIT: Professional Industrial Training
PIL: Professional Industrial Lecture
PSS: Professional Self-Study Course
PMP: Professional Major Project Work

Students joined in 2015 and later

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

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S: Basic Science
E: Basic Engineering
P: Professional Engineering
H: Humanities

B.Tech Curriculum

B.Tech 2014 Batch

Nomenclature

L: Number of Lectures
T: Number of Tutorials
P: Number of Laboratory Sessions
C: Number of Credit

SCY: Basic Sciences-Chemistry
SPH: Basic Sciences-Physics
SMA: Basic Sciences-Mathematics
BET: Basic Engineering Theory
BES: Basic Engineering Practice
PMT: Professional Major Theory
HPF: Humanities Professional Compulsory
SLS: Life Sciences
PML: Professional Major Laboratory
HSS: Humanities and Scoial Sciences Elective
MNS: Minor Stream Elective
PIT: Professional Industrial Training
PIL: Professional Industrial Lecture
PSS: Professional Self-Study Course
PMP: Professional Major Project Work

ME - B.Tech. Mechanical Engineering
2014-Batch

SEMESTER 1

S.No	Course No	Course Name	L	T	P	C	Category
1	CY1002	Chemistry Lab I	1	0	3	3	SCY
2	ID1100	Concepts in Engineering Design	2	0	0	2	BET
3	MA1010	Calculus I Functions of One Variable	3	1	0	4	SMA
4	ME1100	Thermodynamics	3	0	0	3	BET
5	ME1110	Introduction to Mechanical Engineering	2	0	0	2	PMT
6	ME1120	Engineering Drawing	3	0	0	3	BES
7	PH1010	Physics I	3	0	0	3	SPH
8	PH1030	Physics Laboratory I	0	0	3	2	SPH
9	GN1100	Life Skills	2	0	0	0	HPF
		Total Credits :				**24**	

SEMESTER 2

S.No	Course No	Course Name	L	T	P	C	Category
1	AM1100	Engineering Mechanics	4	0	0	4	BET
2	CS1100	Computational Engineering	3	0	0	3	BET
3	CY1001	Chemistry: Structure, Bonding & Reactivity	4	0	0	4	SCY
4	MA1020	Calculus II Functions of Several Variables	3	1	0	4	SMA
5	PH1020	Physics II	3	0	0	3	SPH
6	PH1040	Physics Laboratory II				2	SPH
		Total Credits :				**20**	
	WS1010	Workshop I	0	0	6	4	BES
	WS1020	Workshop II	0	0	3	2	BES

SEMESTER 3

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2200	Strength of Materials	4	0	0	4	PMT
2	BT1010	Life Sciences	2	0	0	2	SLS
3	EE1100	Basic Electrical Engineering	3	0	0	3	BET
4	ID1200	Ecology and Environment	2	0	0	2	BET
5	ME2050	Machine Drawing Practice	1	0	6	5	PML
6	HSE1	Humanities Elective 1	3	0	0	3	HSS
7	MAE1	Mathematics Elective 1	3	0	0	3	SMA
		Total Credits :				**22**	

SEMESTER 4

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2530	Foundations of Fluid Mechanics	3	1	0	4	PMT
2	AM2540	Applied Mechanics Lab	0	0	3	2	PML
3	EE2100	Electrical Sciences	3	0	0	3	PMT
4	ME2220	Kinematics and Dynamics of Machinery	4	0	0	4	PMT
5	ME2240	Instrumentation and Control	4	0	0	4	PMT
6	ME2260	Materials and Design	4	0	0	4	PMT
7	ME2280	Manufacturing Technology	4	0	0	4	PMT
8	MAE4	Mathematics Elective 4	3	0	0	3	SMA
		Total Credits :				**28**	

SEMESTER 5

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3170	Heat Transfer	4	0	0	4	PMT
2	ME3190	Machine Tools & Metrology	4	0	0	4	PMT
3	ME3270	Mechanical Engg Lab I	0	0	3	2	PML
4	ME3310	Turbomachines	4	0	0	4	PMT
5	ME3330	Internal Combustion Engines	3	0	0	3	PMT
6	ME3350	Design of Machine Elements	4	0	0	4	PMT
7	MNS1	Minor Elective 1	3	0	0	3	MNS
		Total Credits :				**24**	

SEMESTER 6

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3250	Refrigeration & Airconditioning	3	0	0	3	PMT
2	ME3260	Thermal Power Engineering	4	0	0	4	PMT
3	ME3280	Mechanical Engineering Lab II	0	0	3	2	PML
4	ME3300	Mechanical Engg. Lab III	0	0	3	2	PML
5	ME3500	Summer Training	0	0	0	2	PIT
6	DPE1	Department Elective 1	3	0	0	3	PMT
7	FRE1	Free Elective 1	3	0	0	3	
8	HSE2	Humanities Elective 2	3	0	0	3	HSS
9	MNS2	Minor Elective 2	3	0	0	3	MNS
		Total Credits :				**25**	

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	IL4020	Industrial Lecture	0	0	0	1	PIL
3	ME4545	Viva voce	0	0	0	2	PMP
4	ME4550	Project I	0	0	0	3	PMP
5	DPE2	Department Elective 2	3	0	0	3	PMT
6	DPE3	Department Elective 3	3	0	0	3	PMT
7	FRE2	Free Elective 2	3	0	0	3	
8	MNS3	Minor Elective 3	3	0	0	3	MNS
9	SSE1	Self-Study Elective 1	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	ME4560	Project II	0	0	0	6	PMP
2	DPE4	Department Elective 4	3	0	0	3	PMT
3	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**12**	

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Req. Cr.	24	20+6	22	28	24	25	23	12	184

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	13	11

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	67	0	3	7	14	10	2	6	0

Remarks

1. Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

2. Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

3. **B.Tech. project**
   - At the end of 6th semester students may opt to do project work or course work
   - Students taking the project option should have satisfactory performance in the mini project in order to continue the major project
   - In lieu of projects, student can do courses for 9 credits (3 credits in 7th semester & 6 credits In 8th semester)
   - Students who fail in mini project shall repeat it or substitute with a course having equivalent credit

4. **B.Tech. Honours**: (Total credit requirement: 184+12 = 196)

   Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.

   1. Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
   2. Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
   3. Project work is a must for B.Tech (Honours) students in B.Tech programme.
   4. Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
   5. Extra credits earned by the students will be counted for prizes and awards.

B.Tech 2015 Batch

Nomenclature

L: Number of lecture hours per week
T: Number of tutorial hours per week
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E: Basic Engineering
P: Professional Engineering
H: Humanities

ME - B.Tech. in Mechanical Engineering
2015 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1100	Life Skills	0	0	0	0	3	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop II	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2220	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2260	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3170	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 2	3	0	0	0	6	9	P
2		Professional elective 3	3	0	0	0	6	9	P
3		Professional elective 4/ Free elective 4	3	0	0	0	6	9	P
4		Professional elective 5/ Free elective 5	3	0	0	0	6	9	P
5		Professional elective 6/ Free elective 6	3	0	0	0	6	9	P
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	0	0	0	0	2	0	H
8		**Credits for semester 7**						**54**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free or professional elective 4 or project*	3	0	0	0	6	9	
2		Free or professional elective 5 or project*	3	0	0	0	6	9	
3		Free or professional elective 6 or project*	3	0	0	0	6	9	
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
6		Ecology and environment	0	0	0	0	2	0	S
		Credits for semester 8						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Note:

(1) Project in other department can be taken only in lieu of Free Electives

(2) Project in ME Department can be taken in lieu of free or departmental electives

(3) It may be noted that any combination of free electives should lead to 72 credits and not necessarily 8 courses

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Credits	49	56+6	52	59	59	52	54	45	432

***Please note that the indicated credits are only for core program.**

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	177+27	27	84	72	432

B.Tech. Curriculum (from July 2015)

Category	Minimum Credit Requirements (CTF)/BOG	ME
Basic Science (S)	84	84
Basic Eng. (E)	45	45
Profession (P)	180	204
Humanities (H)	27	27
Unallocated credits	96	72
Total	**432**	**432**

B.Tech. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	54	204
Humanities (H)	27	27
Unallocated credits	72	72
Total	**162**	**432**
Percentage	**37.5%**	

BTech (honours): (Total credit requirement: 432 + 27 = **459)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

B.Tech 2016 Batch

Nomenclature

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

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S: Basic Science
E: Basic Engineering
P: Professional Engineering
H: Humanities

ME - B.Tech. in Mechanical Engineering
2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1101	Life Skills I	0	0	0	0	2	0	
8	ID1200	Ecology and Environment	2	0	0	0	0	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop II	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2220	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 2	3	0	0	0	6	9	P
2		Professional elective 3	3	0	0	0	6	9	P
3		Professional elective 4/ Free elective 4	3	0	0	0	6	9	P
4		Professional elective 5/ Free elective 5	3	0	0	0	6	9	P
5		Professional elective 6/ Free elective 6	3	0	0	0	6	9	P
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	0	0	0	0	2	0	H
8		**Credits for semester 7**						**54**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free or professional elective 4 or project*	3	0	0	0	6	9	
2		Free or professional elective 5 or project*	3	0	0	0	6	9	
3		Free or professional elective 6 or project*	3	0	0	0	6	9	
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
6		Ecology and environment	0	0	0	0	2	0	S
		Credits for semester 8						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Note:

(1) Project in other department can be taken only in lieu of Free Electives
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B.Tech. - Category wise credit Requirements

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E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

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S: Basic Science
E: Basic Engineering
P: Professional Engineering
H: Humanities

ME - B.Tech. in Mechanical Engineering
2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1101	Life Skills I	0	0	0	0	2	0	
8	ID1200	Ecology and Environment	2	0	0	0	0	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1301	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1302	Workshop II	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 2	3	0	0	0	6	9	P
2		Professional elective 3	3	0	0	0	6	9	P
3		Professional elective 4/ Free elective 4	3	0	0	0	6	9	P
4		Professional elective 5/ Free elective 5	3	0	0	0	6	9	P
5		Professional elective 6/ Free elective 6	3	0	0	0	6	9	P
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	0	0	0	0	2	0	H
8		**Credits for semester 7**						**54**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free or professional elective 4 or project*	3	0	0	0	6	9	
2		Free or professional elective 5 or project*	3	0	0	0	6	9	
3		Free or professional elective 6 or project*	3	0	0	0	6	9	
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
6		Ecology and environment	0	0	0	0	2	0	S
		Credits for semester 8						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Note:

(1) Project in other department can be taken only in lieu of Free Electives
(2) Project in ME Department can be taken in lieu of free or departmental electives
(3) It may be noted that any combination of free electives should lead to 72 credits and not necessarily 8 courses

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Credits	49	56+6	52	59	59	52	54	45	432

***Please note that the indicated credits are only for core program.**

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	177+27	27	84	72	432

B.Tech. Curriculum (from July 2015)

Category	Minimum Credit Requirements (CTF)/BOG	ME
Basic Science (S)	84	84
Basic Eng. (E)	45	45
Profession (P)	180	204
Humanities (H)	27	27
Unallocated credits	96	72
Total	**432**	**432**

B.Tech. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	54	204
Humanities (H)	27	27
Unallocated credits	72	72
Total	**162**	**432**
Percentage	**37.5%**	

BTech (honours): (Total credit requirement: 432 + 27 = **459)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

B.Tech Dual Degree Curriculum

B.Tech DD 2013 Batch

Nomenclature

L: Number of Lectures
T: Number of Tutorials
P: Number of Laboratory Sessions
C: Number of Credit

SCY: Basic Sciences-Chemistry
SPH: Basic Sciences-Physics
SMA: Basic Sciences-Mathematics
BET: Basic Engineering Theory
BES: Basic Engineering Practice
PMT: Professional Major Theory
HPF: Humanities Professional Compulsory
SLS: Life Sciences
PML: Professional Major Laboratory
HSS: Humanities and Scoial Sciences Elective
MNS: Minor Stream Elective
PIT: Professional Industrial Training
PIL: Professional Industrial Lecture
PSS: Professional Self-Study Course
PMP: Professional Major Project Work

BRANCH CODE: ME25

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

2013-Batch

SEMESTER 1

S.No	Course No	Course Name	L	T	P	C	Category
1	CY1002	Chemistry Lab I	1	0	3	3	SCY
2	ID1100	Concepts in Engineering Design	2	0	0	2	BET
3	MA1010	Calculus I Functions of One Variable	3	1	0	4	SMA
4	ME1100	Thermodynamics	3	0	0	3	BET
5	ME1110	Introduction to Mechanical Engineering	2	0	0	2	PMT
6	ME1120	Engineering Drawing	3	0	0	3	BES
7	PH1010	Physics I	3	0	0	3	SPH
8	PH1030	Physics Laboratory I	0	0	3	2	SPH
9	GN1100	Life Skills	2	0	0	2	HPF
		Total Credits :				**24**	

SEMESTER 2

S.No	Course No	Course Name	L	T	P	C	Category
1	AM1100	Engineering Mechanics	4	0	0	4	BET
2	CS1100	Computational Engineering	3	0	0	3	BET
3	CY1001	Chemistry: Structure, Bonding & Reactivity	4	0	0	4	SCY
4	MA1020	Calculus II Functions of Several Variables	3	1	0	4	SMA
5	PH1020	Physics II	3	0	0	3	SPH
6	PH1040	Physics Laboratory II				2	SPH
		Total Credits :				**20**	
	WS1010	Workshop I	0	0	6	4	BES
	WS1020	Workshop II	0	0	3	2	BES

SEMESTER 3

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2200	Strength of Materials	4	0	0	4	PMT
2	BT1010	Life Sciences	2	0	0	2	SLS
3	EE1100	Basic Electrical Engineering	3	0	0	3	BET
4	ID1200	Ecology and Environment	2	0	0	2	BET
5	ME2050	Machine Drawing Practice	1	0	6	5	PML
6	HSE1	Humanities Elective 1	3	0	0	3	HSS
7	MAE1	Mathematics Elective 1	3	0	0	3	SMA
		Total Credits :				**22**	

SEMESTER 4

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2530	Foundations of Fluid Mechanics	3	1	0	4	PMT
2	AM2540	Applied Mechanics Lab	0	0	3	2	PML
3	EN2100	Electrical Sciences	3	0	0	3	PMT
4	ME2220	Kinematics and Dynamics of Machinery	4	0	0	4	PMT
5	ME2240	Instrumentation and Control	4	0	0	4	PMT
6	ME2260	Materials and Design	4	0	0	4	PMT
7	ME2280	Manufacturing Technology	4	0	0	4	PMT
8	MAE4	Mathematics Elective 4	3	0	0	3	SMA
		Total Credits :				**28**	

SEMESTER 5

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3170	Heat Transfer	4	0	0	4	PMT
2	ME3190	Machine Tools & Metrology	4	0	0	4	PMT
3	ME3270	Mechanical Engg Lab I	0	0	3	2	PML
4	ME3310	Turbomachines	4	0	0	4	PMT
5	ME3330	Internal Combustion Engines	3	0	0	3	PMT
6	ME3350	Design of Machine Elements	4	0	0	4	PMT
7	MNS1	Minor Elective 1	3	0	0	3	MNS
		Total Credits :				**24**	

SEMESTER 6

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3250	Refrigeration & Airconditioning	3	0	0	3	PMT
2	ME3260	Thermal Power Engineering	4	0	0	4	PMT
3	ME3280	Mechanical Engineering Lab II	0	0	3	2	PML
4	ME3300	Mechanical Engg. Lab III	0	0	3	2	PML
5	ME3500	Summer Training	0	0	0	2	PIT
6	DPE1	Department Elective 1	3	0	0	3	PMT
7	FRE1	Free Elective 1	3	0	0	3	
8	HSE2	Humanities Elective 2	3	0	0	3	HSS
9	MNS2	Minor Elective 2	3	0	0	3	MNS
		Total Credits :				**25**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Product Design

2013-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6800	Finite Element Analysis	3	0	0	3	PMT
3	ME6820	Fundamentals of Engineering Design	3	0	0	3	PMT
4	ME6830	Principles of Product Design	3	0	0	3	PMT
5	ME6840	Design for Manufacture and Assembly	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6780	Design Synthesis	3	0	0	3	PMT
3	ME6850	Product Reliability	3	0	0	3	PMT
4	ME6870	CAD/CAM for Product Design	3	0	0	3	PMT
5	ME6880	Product Design Lab	0	0	3	2	PML
6	ME7400	Mechatronic Systems	3	0	0	3	PMT
7	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
8	HSE4	Humanities Elective 4	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
3	DDE2	Dual Degree Elective 3	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

- Any two PMT electives can be taken as Pass/Fail courses

- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Intelligent Manufacturing
2013-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME7010	Microprocessors in Automation	3	0	0	3	PMT
3	ME7020	Robotics and Robot Applications	3	0	0	3	PMT
4	ME7050	CNC and Adaptive Control	3	0	0	3	PMT
5	ME7240	Modelling and Simulation in Manufacturing	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective 1	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME7120	Sensors for Intelligent Manufacturing	3	0	0	3	PMT
3	ME7130	Computer Int. Manuf. Lab I	0	0	3	2	PML
4	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
5	ME7250	Rapid Response Manufacturing	3	0	0	3	PMT
6	ME7400	Mechatronic Systems	3	0	0	3	PMT
7	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
3	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.
- Any two PMT electives can be taken as Pass/Fail courses
- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.
- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Thermal Engineering

2013-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6041	Incompressible Fluid Flow	3	0	0	3	PMT
3	ME6140	Applied Thermodynamics	3	0	0	3	PMT
4	ME6280	Design & Optimisation of Energy Systems	3	0	0	3	PMT
5	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6080	Measurement in Thermal Engg	4	0	0	4	PMT
3	ME6150	Numerical Methods in Thermal Engg	4	0	0	4	PMT
4	ME6290	Advanced Energy Conversion	3	0	0	3	PMT
6	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
7	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
3	DDE5	Dual Degree Elective 5	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	13	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	87	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

- Any two PMT electives can be taken as Pass/Fail courses

- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

B.Tech DD 2014 Batch

Nomenclature

L: Number of Lectures
T: Number of Tutorials
P: Number of Laboratory Sessions
C: Number of Credit

SCY: Basic Sciences-Chemistry
SPH: Basic Sciences-Physics
SMA: Basic Sciences-Mathematics
BET: Basic Engineering Theory
BES: Basic Engineering Practice
PMT: Professional Major Theory
HPF: Humanities Professional Compulsory
SLS: Life Sciences
PML: Professional Major Laboratory
HSS: Humanities and Scoial Sciences Elective
MNS: Minor Stream Elective
PIT: Professional Industrial Training
PIL: Professional Industrial Lecture
PSS: Professional Self-Study Course
PMP: Professional Major Project Work

BRANCH CODE: ME25

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

2014-Batch

SEMESTER 1

S.No	Course No	Course Name	L	T	P	C	Category
1	CY1002	Chemistry Lab I	1	0	3	3	SCY
2	ID1100	Concepts in Engineering Design	2	0	0	2	BET
3	MA1010	Calculus I Functions of One Variable	3	1	0	4	SMA
4	ME1100	Thermodynamics	3	0	0	3	BET
5	ME1110	Introduction to Mechanical Engineering	2	0	0	2	PMT
6	ME1120	Engineering Drawing	3	0	0	3	BES
7	PH1010	Physics I	3	0	0	3	SPH
8	PH1030	Physics Laboratory I	0	0	3	2	SPH
9	GN1100	Life Skills	2	0	0	2	HPF
		Total Credits :				**24**	

SEMESTER 2

S.No	Course No	Course Name	L	T	P	C	Category
1	AM1100	Engineering Mechanics	4	0	0	4	BET
2	CS1100	Computational Engineering	3	0	0	3	BET
3	CY1001	Chemistry: Structure, Bonding & Reactivity	4	0	0	4	SCY
4	MA1020	Calculus II Functions of Several Variables	3	1	0	4	SMA
5	PH1020	Physics II	3	0	0	3	SPH
6	PH1040	Physics Laboratory II				2	SPH
		Total Credits :				**20**	
	WS1010	Workshop I	0	0	6	4	BES
	WS1020	Workshop II	0	0	3	2	BES

SEMESTER 3

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2200	Strength of Materials	4	0	0	4	PMT
2	BT1010	Life Sciences	2	0	0	2	SLS
3	EE1100	Basic Electrical Engineering	3	0	0	3	BET
4	ID1200	Ecology and Environment	2	0	0	2	BET
5	ME2050	Machine Drawing Practice	1	0	6	5	PML
6	HSE1	Humanities Elective 1	3	0	0	3	HSS
7	MAE1	Mathematics Elective 1	3	0	0	3	SMA
		Total Credits :				**22**	

SEMESTER 4

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2530	Foundations of Fluid Mechanics	3	1	0	4	PMT
2	AM2540	Applied Mechanics Lab	0	0	3	2	PML
3	EN2100	Electrical Sciences	3	0	0	3	PMT
4	ME2220	Kinematics and Dynamics of Machinery	4	0	0	4	PMT
5	ME2240	Instrumentation and Control	4	0	0	4	PMT
6	ME2260	Materials and Design	4	0	0	4	PMT
7	ME2280	Manufacturing Technology	4	0	0	4	PMT
8	MAE4	Mathematics Elective 4	3	0	0	3	SMA
		Total Credits :				**28**	

SEMESTER 5

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3170	Heat Transfer	4	0	0	4	PMT
2	ME3190	Machine Tools & Metrology	4	0	0	4	PMT
3	ME3270	Mechanical Engg Lab I	0	0	3	2	PML
4	ME3310	Turbomachines	4	0	0	4	PMT
5	ME3330	Internal Combustion Engines	3	0	0	3	PMT
6	ME3350	Design of Machine Elements	4	0	0	4	PMT
7	MNS1	Minor Elective 1	3	0	0	3	MNS
		Total Credits :				**24**	

SEMESTER 6

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3250	Refrigeration & Airconditioning	3	0	0	3	PMT
2	ME3260	Thermal Power Engineering	4	0	0	4	PMT
3	ME3280	Mechanical Engineering Lab II	0	0	3	2	PML
4	ME3300	Mechanical Engg. Lab III	0	0	3	2	PML
5	ME3500	Summer Training	0	0	0	2	PIT
6	DPE1	Department Elective 1	3	0	0	3	PMT
7	FRE1	Free Elective 1	3	0	0	3	
8	HSE2	Humanities Elective 2	3	0	0	3	HSS
9	MNS2	Minor Elective 2	3	0	0	3	MNS
		Total Credits :				**25**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Product Design

2014-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6800	Finite Element Analysis	3	0	0	3	PMT
3	ME6820	Fundamentals of Engineering Design	3	0	0	3	PMT
4	ME6830	Principles of Product Design	3	0	0	3	PMT
5	ME6840	Design for Manufacture and Assembly	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6224	Design Synthesis	3	0	0	3	PMT
3	ME6226	Product Reliability	3	0	0	3	PMT
4	ME6228	CAD/CAM for Product Design	3	0	0	3	PMT
5	ME6880	Product Design Lab	0	0	3	2	PML
6	ME6332	Mechatronic Systems	3	0	0	3	PMT
7	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
8	HSE4	Humanities Elective 4	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
3	DDE2	Dual Degree Elective 3	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

- Any two PMT electives can be taken as Pass/Fail courses

- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Intelligent Manufacturing
2014-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME7010	Microprocessors in Automation	3	0	0	3	PMT
3	ME7020	Robotics and Robot Applications	3	0	0	3	PMT
4	ME7050	CNC and Adaptive Control	3	0	0	3	PMT
5	ME5307	Modelling and Simulation in Manufacturing	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective 1	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME5302	Sensors for Intelligent Manufacturing	3	0	0	3	PMT
3	ME7130	Computer Int. Manuf. Lab I	0	0	3	2	PML
4	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
5	ME7250	Rapid Response Manufacturing	3	0	0	3	PMT
6	ME7400	Mechatronic Systems	3	0	0	3	PMT
7	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
3	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

- Any two PMT electives can be taken as Pass/Fail courses

- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Thermal Engineering

2014-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6040	Incompressible Fluid Flow	3	0	0	3	PMT
3	ME5105	Applied Thermodynamics	3	0	0	3	PMT
4	ME6280	Design & Optimisation of Energy Systems	3	0	0	3	PMT
5	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6080	Measurement in Thermal Engg	4	0	0	4	PMT
3	ME5107	Numerical Methods in Thermal Engg	4	0	0	4	PMT
4	ME6150	Advanced Energy Conversion	3	0	0	3	PMT
6	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
7	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
3	DDE5	Dual Degree Elective 5	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	13	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	87	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

- Any two PMT electives can be taken as Pass/Fail courses

- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

B.Tech DD 2015 Batch

Nomenclature

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

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S: Basic Science
E: Basic Engineering
P: Professional Engineering
H: Humanities

Branch Code: ME25
Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering
2015-Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1100	Life Skills	0	0	0	0	3	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop I	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honours elective 1	3	0	0	0	6	9	
		Credits for semester 6 for Honours						**61**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Mechanical Design
2015-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional Elective 2	3	0	0	0	6	9	P
2	ME5205	Theory of Vibration	3	0	0	0	6	9	P
3	ME5207	Design with Advanced Materials	3	0	0	0	6	9	P
4		Free Elective 4	3	0	0	0	6	9	
5		Free Elective 5	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	2	0	0	0	0	0	H
8	ME5281	Mechanical Design Laboratory	0	0	0	3	0	3	P
		Credits for semester 7						**57**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**66**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME5203	Advanced Mechanics of Solids	3	0	0	0	6	9	P
2		Professional Elective 3	3	0	0	0	6	9	P
3		Professional Elective 4	3	0	0	0	6	9	P
4		Free Elective 6	3	0	0	0	6	9	
5		Free Elective 7	3	0	0	0	6	9	
6	ME5280	Design Practice with CAD Tools	1	0	0	3	2	6	P
		Credits for semester 8						**51**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**60**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490#	Project phase 1	0	0	0	0	22	22	
		Credits for summer						**22**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490+	Project phase 2	0	0	0	0	23	23	
2	ME	Professional Elective 5	3	0	0	0	6	9	P
3		Free Elective 8	3	0	0	0	6	9	
4	ME5204	Finite Element Analysis	3	0	0	0	6	9	P
		Credits for semester 9						**50**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	57	51	22	50	40	553

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**168+72+85**	**27**	**84**	**72**	**553**

DD Curriculum Mechanical Design

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	240
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**553**

DD Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	240
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**468 + 85**
Percentage		

Note:

- **Professional Elective 1 - to be chosen from 4000 level;**
- **Professional Elective 2-4 - to be chosen from 5000 + level.**

BTech (honours) + M.Tech: (Total credit requirement: 553 + 27 = **580)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No.31 & 32)

STREAM: Intelligent Manufacturing
2015-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7240	Modeling and Simulation in Manufacturing	3	0	0	0	6	9	P
2	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9	P
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9	P
4	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9	P
5		Free Elective 4	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	0	0	0	0	2	0	H
		Credits for semester 7						**54**	
	ME	**Honours elective 2**	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free Elective 5	3	0	0	0	6	9	
2		Free Elective 6	3	0	0	0	6	9	
3	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9	P
4	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9	P
5	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4	P
6	ME	Professional Elective 2	3	0	0	0	6	9	P
		Credits for semester 7						**49**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**58**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2	ME	Professional Elective 3	3	0	0	0	6	9	P
3	ME	Professional Elective 4	3	0	0	0	6	9	P
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	54	49	21	60	40	557

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	172+72+85	27	84	72	557

DD Curriculum Intelligent Manufacturing

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	244
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	244
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**472 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 557 + 27 = **584)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Thermal Engineering

2015-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6080	Measurement in Thermal Engg	2	0	0	3	5	10	P
2	ME6150	Numerical Methods in Thermal Engg	2	0	0	3	5	10	P
3		Professional elective 2	3	0	0	0	6	9	P
4		Free elective 4	3	0	0	0	6	9	
5		Free elective 5	3	0	0	0	6	9	
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	2	0	0	0	2	0	H
8		**Credits for semester 7**						**56**	
		Honours elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**65**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 3	3	0	0	0	6	9	P
2		Professional elective 4	3	0	0	0	6	9	P
3		Professional elective 5	3	0	0	0	6	9	P
4		Professional elective 6	3	0	0	0	6	9	P
5		Professional elective 7	3	0	0	0	6	9	P
6		**Credits for semester 8**						**45**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**54**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2		Free elective 6	3	0	0	0	6	9	
3		Free elective 7	3	0	0	0	6	9	
4		Free elective 8	3	0	0	0	6	9	
5		Professional elective 8	3	0	0	0	6	9	P
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	56	45	21	60	40	555

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**170+72+85**	**27**	**84**	**72**	**555**

DD Curriculum Thermal stream

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	242
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	242
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**470 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 555 + 27 = **582)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

B.Tech DD 2016 Batch

Nomenclature

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

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S: Basic Science
E: Basic Engineering
P: Professional Engineering
H: Humanities

Branch Code: ME25
Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering
2016-Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8	GN1101	Life Skills I	0	0	0	0	2	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1120	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop I	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Mechanical Design
2016-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional Elective 2	3	0	0	0	6	9	P
2	ME5205	Theory of Vibration	3	0	0	0	6	9	P
3	ME5207	Design with Advanced Materials	3	0	0	0	6	9	P
4		Free Elective 4	3	0	0	0	6	9	
5		Free Elective 5	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	2	0	0	0	0	0	H
8	ME5281	Mechanical Design Laboratory	0	0	0	3	0	3	P
		Credits for semester 7						**57**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**66**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME5203	Advanced Mechanics of Solids	3	0	0	0	6	9	P
2		Professional Elective 3	3	0	0	0	6	9	P
3		Professional Elective 4	3	0	0	0	6	9	P
4		Free Elective 6	3	0	0	0	6	9	
5		Free Elective 7	3	0	0	0	6	9	
6	ME5280	Design Practice with CAD Tools	1	0	0	3	2	6	P
		Credits for semester 8						**51**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**60**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490#	Project phase 1	0	0	0	0	22	22	
		Credits for summer						**22**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490+	Project phase 2	0	0	0	0	23	23	
2	ME	Professional Elective 5	3	0	0	0	6	9	P
3		Free Elective 8	3	0	0	0	6	9	
4	ME5204	Finite Element Analysis	3	0	0	0	6	9	P
		Credits for semester 9						**50**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	57	51	22	50	40	553

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**168+72+85**	**27**	**84**	**72**	**553**

DD Curriculum Mechanical Design

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	240
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**553**

DD Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	240
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**468 + 85**
Percentage		

Note:

- **Professional Elective 1 - to be chosen from 4000 level;**
- **Professional Elective 2-4 - to be chosen from 5000 + level.**

BTech (honours) + M.Tech: (Total credit requirement: 553 + 27 = **580)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No.31 & 32)

STREAM: Intelligent Manufacturing
2016-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7240	Modeling and Simulation in Manufacturing	3	0	0	0	6	9	P
2	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9	P
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9	P
4	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9	P
5		Free Elective 4	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	0	0	0	0	2	0	H
		Credits for semester 7						**54**	
	ME	**Honours elective 2**	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free Elective 5	3	0	0	0	6	9	
2		Free Elective 6	3	0	0	0	6	9	
3	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9	P
4	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9	P
5	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4	P
6	ME	Professional Elective 2	3	0	0	0	6	9	P
		Credits for semester 7						**49**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**58**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2	ME	Professional Elective 3	3	0	0	0	6	9	P
3	ME	Professional Elective 4	3	0	0	0	6	9	P
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	54	49	21	60	40	557

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	172+72+85	27	84	72	557

DD Curriculum Intelligent Manufacturing

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	244
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	244
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**472 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 557 + 27 = **584)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Thermal Engineering
2016-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6080	Measurement in Thermal Engg	2	0	0	3	5	10	P
2	ME6150	Numerical Methods in Thermal Engg	2	0	0	3	5	10	P
3		Professional elective 2	3	0	0	0	6	9	P
4		Free elective 4	3	0	0	0	6	9	
5		Free elective 5	3	0	0	0	6	9	
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	2	0	0	0	2	0	H
8		**Credits for semester 7**						**56**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**65**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective3	3	0	0	0	6	9	P
2		Professional elective 4	3	0	0	0	6	9	P
3		Professional elective 5	3	0	0	0	6	9	P
4		Professional elective 6	3	0	0	0	6	9	P
5		Professional elective 7	3	0	0	0	6	9	P
6		**Credits for semester 8**						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2		Free elective 6	3	0	0	0	6	9	
3		Free elective 7	3	0	0	0	6	9	
4		Free elective 8	3	0	0	0	6	9	
5		Professional elective 8	3	0	0	0	6	9	P
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	56	45	21	60	40	555

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**170+72+85**	**27**	**84**	**72**	**555**

DD Curriculum Thermal stream

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	242
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**555**

DD. - Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	242
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**470 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 555 + 27 = **582)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

B.Tech DD 2017 Batch

Nomenclature

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

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S: Basic Science
E: Basic Engineering
P: Professional Engineering
H: Humanities

Branch Code: ME25
Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering
2017-Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8	GN1101	Life Skills I	0	0	0	0	2	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop I	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Mechanical Design
2017-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional Elective 2	3	0	0	0	6	9	P
2	ME5205	Theory of Vibration	3	0	0	0	6	9	P
3	ME5207	Design with Advanced Materials	3	0	0	0	6	9	P
4		Free Elective 4	3	0	0	0	6	9	
5		Free Elective 5	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	2	0	0	0	0	0	H
8	ME5281	Mechanical Design Laboratory	0	0	0	3	0	3	P
		Credits for semester 7						**57**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**66**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME5203	Advanced Mechanics of Solids	3	0	0	0	6	9	P
2		Professional Elective 3	3	0	0	0	6	9	P
3		Professional Elective 4	3	0	0	0	6	9	P
4		Free Elective 6	3	0	0	0	6	9	
5		Free Elective 7	3	0	0	0	6	9	
6	ME5280	Design Practice with CAD Tools	1	0	0	3	2	6	P
		Credits for semester 8						**51**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**60**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490#	Project phase 1	0	0	0	0	22	22	
		Credits for summer						**22**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490+	Project phase 2	0	0	0	0	23	23	
2	ME	Professional Elective 5	3	0	0	0	6	9	P
3		Free Elective 8	3	0	0	0	6	9	
4	ME5204	Finite Element Analysis	3	0	0	0	6	9	P
		Credits for semester 9						**50**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	57	51	22	50	40	553

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**168+72+85**	**27**	**84**	**72**	**553**

DD Curriculum Mechanical Design

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	240
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**553**

DD Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	240
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**468 + 85**
Percentage		

Note:

- **Professional Elective 1 - to be chosen from 4000 level;**
- **Professional Elective 2-4 - to be chosen from 5000 + level.**

BTech (honours) + M.Tech: (Total credit requirement: 553 + 27 = **580)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No.31 & 32)

STREAM: Intelligent Manufacturing
2017-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7240	Modeling and Simulation in Manufacturing	3	0	0	0	6	9	P
2	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9	P
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9	P
4	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9	P
5		Free Elective 4	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	0	0	0	0	2	0	H
		Credits for semester 7						**54**	
	ME	**Honours elective 2**	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free Elective 5	3	0	0	0	6	9	
2		Free Elective 6	3	0	0	0	6	9	
3	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9	P
4	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9	P
5	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4	P
6	ME	Professional Elective 2	3	0	0	0	6	9	P
		Credits for semester 7						**49**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**58**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2	ME	Professional Elective 3	3	0	0	0	6	9	P
3	ME	Professional Elective 4	3	0	0	0	6	9	P
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	54	49	21	60	40	557

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	172+72+85	27	84	72	557

DD Curriculum Intelligent Manufacturing

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	244
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	244
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**472 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 557 + 27 = **584)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Thermal Engineering
2017-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6080	Measurement in Thermal Engg	2	0	0	3	5	10	P
2	ME6150	Numerical Methods in Thermal Engg	2	0	0	3	5	10	P
3		Professional elective 2	3	0	0	0	6	9	P
4		Free elective 4	3	0	0	0	6	9	
5		Free elective 5	3	0	0	0	6	9	
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	2	0	0	0	2	0	H
8		**Credits for semester 7**						**56**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**65**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective3	3	0	0	0	6	9	P
2		Professional elective 4	3	0	0	0	6	9	P
3		Professional elective 5	3	0	0	0	6	9	P
4		Professional elective 6	3	0	0	0	6	9	P
5		Professional elective 7	3	0	0	0	6	9	P
6		**Credits for semester 8**						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2		Free elective 6	3	0	0	0	6	9	
3		Free elective 7	3	0	0	0	6	9	
4		Free elective 8	3	0	0	0	6	9	
5		Professional elective 8	3	0	0	0	6	9	P
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	56	45	21	60	40	555

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**170+72+85**	**27**	**84**	**72**	**555**

DD Curriculum Thermal stream

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	242
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**555**

DD. - Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	242
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**470 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 555 + 27 = **582)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

M.Tech Curriculum

M.Tech 2016 Batch

Nomenclature

L: Number of Lectures
T: Number of Tutorials
P: Number of Laboratory Sessions
C: Number of Credit

Branch Code: ME1

M.Tech. in MECHANICAL ENGINEERING

STREAM: Thermal Engineering

2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6010	Advanced Heat and Mass Transfer	3	0	0	0	6	9
2	ME6020	I.C. Engines Combustion and Pollution	3	0	0	0	6	9
3	ME6030	Refrigeration and Cryogenics	3	0	0	0	6	9
4	ME6040	Incompressible Fluid Flow	3	0	0	0	6	9
5	ME6050	Principles of Turbomachinery	3	0	0	0	6	9
6	ME6140	Applied Thermodynamics	3	0	0	0	6	9
		Total Credits :						**54**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6150	Numerical Methods in Thermal Engineering	2	0	0	3	4	9
2	ME6080	Measurements in Thermal Engineering	2	0	0	3	4	9
3	ME6290	Advanced Energy Conversion	3	0	0	0	6	9
4	ME	Professional Elective I	3	0	0	0	6	9
5	ME	Professional Elective II	3	0	0	0	6	9
6	ME6070	Thermal Engg. Lab	0	0	0	3	0	3
		Total Credits :						**48**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20*
		Total Credits						**20**

** Project (ME7990 *) grade assigned in 4th semester*

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	25	25*
2	ME	Professional Elective III/Free elective	3	0	0	0	6	9
3	ME	Professional Elective IV	3	0	0	0	6	9
4		**Total Credits :**						**43**

** Project (ME7990+) grade assigned in 4th semester*

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	54	48	20	43	40	205

* Free Elective can be taken either in 2nd or 3rd semester

* Grade assigned in fourth semester

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5520	Explosion and Safety	3	0	0	0	6	9
2	ME5530	Thermofluid dynamics of climate	3	0	0	0	6	9
3	ME5550	Flow and Thermal Instabilities	3	0	0	0	6	9
4	ME5560	Heat and Mass Transfer in Biological Systems	3	0	0	0	6	9
5	ME5570	Jetflow & Acoustics	3	0	0	0	6	9
6	ME6001	Theory of Fire Propagation	3	0	0	0	6	9
7	ME6002	Turbomachinery Noise and Control	3	0	0	0	6	9
8	ME6060	Fundamentals of Combustion	3	0	0	0	6	9
9	ME6100	Rocket technology	3	0	0	0	6	9
10	ME6110	Combustion technology	3	0	0	0	6	9
11	ME6120	Air-breathing engines	3	0	0	0	6	9
12	ME6130	Transport Phenomena	3	0	0	0	6	9
13	ME6180	Energy and Environment	3	0	0	0	6	9
14	ME6200	Conduction & Radiation	3	0	0	0	6	9
15	ME6220	Heat exchanger design	3	0	0	0	6	9
16	ME6230	Convection and two-phase flow	3	0	0	0	6	9
17	ME6250	Heat Transfer in Energy Systems	3	0	0	0	6	9
18	ME6270	Design of Power Plant Systems	3	0	0	0	6	9
19	ME6280	Design and Optimization of Energy Systems	3	0	0	0	6	9
20	ME6300	Design of hydro-turbines	3	0	0	0	6	9
21	ME6310	Design of pumps	3	0	0	0	6	9
22	ME6320	Pump Application Engg.	3	0	0	0	6	9
23	ME6330	Cavitation	3	0	0	0	6	9
24	ME6340	Vortex Element Methods	3	0	0	0	6	9
25	ME6350	Mechanical Design of Hydroturbomachines	3	0	0	0	6	9
26	ME6360	Standards in Hydroturbomachines	3	0	0	0	6	9
27	ME6370	Experimental Methods in Hydroturbomachines	3	0	0	0	6	9
28	ME6380	Design of Turbomachines	3	0	0	0	6	9
29	ME6390	Micro Hydro Power	3	0	0	0	6	9
30	ME6400	Design of combustion engines	3	0	0	0	6	9
31	ME6410	Two Stroke Engines	3	0	0	0	6	9
32	ME6420	Simulation of IC Engines Processes	3	0	0	0	6	9
33	ME6430	Engine systems and performance	3	0	0	0	6	9
34	ME6440	Alternative Fuels	3	0	0	0	6	9
35	ME6450	Gas Turbines	3	0	0	0	6	9
36	ME6460	CFD and Its Applications to Engine Processes	3	0	0	0	6	9
37	ME6470	Engine Instrumentation & Electronic Management	3	0	0	0	6	9
38	ME6480	Transport Processes in Engines	3	0	0	0	6	9
39	ME6490	Laser Diagnostics in Engines	3	0	0	0	6	9
40	ME6500	Airconditioning and ventilation	3	0	0	0	6	9
41	ME6510	Refrigeration machinery and components	3	0	0	0	6	9
42	ME6520	Sorption refrigeration and heating systems	3	0	0	0	6	9
43	ME6530	HVAC Systems and Applications	3	0	0	0	6	9
44	ME6540	Food Processing, Storage and Transport	3	0	0	0	6	9
45	ME6550	Vacuum Engineering	3	0	0	0	6	9
46	ME6560	Advanced Cryogenic Systems	3	0	0	0	6	9
47	ME6570	Thermal Energy Conservation	3	0	0	0	6	9
48	ME6580	Utilization of Solar Energy	3	0	0	0	6	9
49	ME6590	Renewable Energy Technology	3	0	0	0	6	9
50	ME6600	Aerodynamic Design of Axial Compressors and Turbines	3	0	0	0	6	9

S.No	Course No	Course Name	L	T	E	P	O	C
51	ME6610	Theory of steam and gas turbines	3	0	0	0	6	9
52	ME6620	Theory and design of centrifugal m/c	3	0	0	0	6	9
53	ME6630	Theory of axial compressors	3	0	0	0	6	9
54	ME6640	Advanced Topics in Turbomachinery	3	0	0	0	6	9
55	ME6650	Computational Fluid Dynamics of Turbomachinery	3	0	0	0	6	9
56	ME6660	Fans, Blowers and Compressors	3	0	0	0	6	9
57	ME6670	Gas Turbine Engineering	3	0	0	0	6	9
58	ME6680	Measurement Techniques in Thermal Turbomachines	3	0	0	0	6	9
59	ME6690	Compact Heat Exchangers	3	0	0	0	6	9
60	ME6910	Diagnostic Methods in Combustion Systems	3	0	0	0	6	9
61	ME6920	Performance Analysis of Hydroturbomachines	3	0	0	0	6	9
62	ME6930	Engine Tribology	3	0	0	0	6	9
63	ME6940	Engine Noise and Vibrations	3	0	0	0	6	9
64	ME7270	Microscale Fluid Flow and Machinery	3	0	0	0	6	9
65	ME7770	Theory & Tech. of Fuel Cells	3	0	0	0	6	9
66	ME7790	Heat and Mass Transfer in Porous Media	3	0	0	0	6	9

Branch Code: ME2

M.Tech. in MECHANICAL ENGINEERING

STREAM: DESIGN

2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6000	Computational Methods in Engineering	3	1	0	0	6	10
2	ME6700	Advanced Mechanics of Solids	3	0	0	0	6	9
3	ME6830	Principles of Product Design	3	0	0	0	6	9
4	ME7360	Theory of Vibration	3	0	0	0	6	9
5	ME7390	Design with Advanced Materials	3	0	0	0	6	9
6	ME6790	CAD and Machine Elements Laboratory	0	0	0	3	0	3
		Total Credits :						**49**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6800	Finite Element Analysis	3	0	0	0	6	9
2	ME7700	Design Practice	1	0	0	3	2	6
3	ME6730	Mechanical Design Lab	0	0	0	3	0	3
4		Professional Elective I	3	0	0	0	6	9
5		Professional Elective II	3	0	0	0	6	9
6		Professional Elective III	3	0	0	0	6	9
7		Professional Elective IV	3	0	0	0	6	9
		Total Credits :						**54**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20*
		Total Credits						**20**

** Project (ME7990 *) grade assigned in 4th semester*

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990+	Project Phase II	0	0	0	0	25	25*
2		Professional Elective V	3	0	0	0	6	9
3		Professional Elective VI	3	0	0	0	6	9
		Total Credits :						**43**

** Project (ME7990+) grade assigned in 4th semester*

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	49	54	20	43	40	206

* Grade assigned in fourth semester

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1	CT7000	Composite Materials Science	3	0	0	0	6	9
2	CT7120	Modeling and Equipment Des for Comp Proc	3	0	0	0	6	9
3	ID5020	Multi-body Dynamics and Applications	3	0	0	0	6	9
4	ID6010	Constitutive Modeling in Continuum Mechanics	3	0	0	0	6	9
5	ID7010	Advanced Finite Element Analysis	3	0	0	0	6	9
6	ME6750	Gear Design	3	0	0	0	6	9
7	ME6760	Design of Mechanical Transmission Elements	3	0	0	0	6	9
8	ME6770	Design of Pressure vessels and Piping	3	0	0	0	6	9
9	ME6780	Design Synthesis	3	0	0	0	6	9
10	ME6810	Transmission Mechanisms and Manipulators	3	0	0	0	6	9
11	ME6840	Design for Manufacture and Assembly	3	0	0	0	6	9
12	ME6850	Product Reliability	3	0	0	0	6	9
13	ME6870	CAD/CAM for product design	3	0	0	0	6	9
14	ME7020	Robotics and Robot applications	3	0	0	0	6	9
15	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9
16	ME7300	Friction and Wear in Machinery	3	0	0	0	6	9
17	ME7400	Mechatronic Systems	3	0	0	0	6	9
18	ME7430	Oil Hydraulics and Pneumatic Systems	3	0	0	0	6	9
19	ME7470	Industrial Instrumentation	3	0	0	0	6	9
20	ME7500	Measurement Systems	3	0	0	0	6	9
21	ME7640	Tribo Design and Analysis	3	0	0	0	6	9
22	ME7660	Nonlinear Solid Mechanics	3	0	0	0	6	9
23	ME7680	Optimization Methods for Mechanical Design	3	0	0	0	6	9
24	ME7710	Advanced Vibration and Acoustics	3	0	0	0	6	9
25	ME7740	Structural Health and Integrity Monitoring	3	0	0	0	6	9
26	ME7820	Rotor Dynamics	3	0	0	0	6	9
27	ME7830	Random Vibrations	3	0	0	0	6	9
28	ME7840	Signal Processing in Mechanical systems	3	0	0	0	6	9
29	ME7850	Modal Analysis of Mechanical systems	3	0	0	0	6	9
30	ME7860	Tribo-Instrumentation	3	0	0	0	6	9
31	ME7870	Diagnostic Maintenance	3	0	0	0	6	9
32	ME7880	Vehicular Vibration	3	0	0	0	6	9
33	ME7890	Advanced Applied Finite Element	3	0	0	0	6	9
34	ME7910	Acoustics and Noise Control	3	0	0	0	6	9
35	ME7920	Applied Finite Element	3	0	0	0	6	9
36	ME7930	Chaotic Vibrations	3	0	0	0	6	9
37	ME 7190	Introduction to Fracture Mechanics	3	0	0	0	6	9
38	ME6012	Mechanics of Human Movement	3	0	0	0	6	9
39	ME6710	Theory of Mechanisms	3	0	0	0	6	9
40	ME6720	Failure Analysis and Design	3	0	0	0	6	9
41	ME6016	Mechanics of Thin Films for Microsystem Design	3	0	0	0	6	9
42	ME6003	Variational Principles in Mechanics	3	0	0	0	6	9
43	ME6820	Fundamentals of Engineering Design	3	0	0	0	6	9
44	ME6015	Elastic waves and ultrasonics	3	0	0	0	6	9
45	ID6070	Mechanics of Viscoelastic materials	3	0	0	0	6	9
46	ME8001	Mechanics of Mixtures	3	0	0	0	6	9
47	ME7023	Foundations of computational materials modelling	3	0	0	0	6	9

Branch Code: ME3

M.Tech. in MECHANICAL ENGINEERING
STREAM: MANUFACTURING ENGINEERING
2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7090	Advanced Materials and Processing	3	0	0	0	6	9
2	ME6000	Computational Methods in Engineering	3	1	0	0	6	10
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9
4	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9
5	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9
		Total Credits :						**46**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7030	Metrology and Computer Aided Inspection	3	0	0	0	6	9
2	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4
3	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9
4	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9
5		Professional Elective I	3	0	0	0	6	9
6		Professional Elective II	3	0	0	0	6	9
		Total Credits :						**49**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20*
		Total Credits						**20**

** Project (ME7990 *) grade assigned in 4th semester*

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	25	25*
2		Professional Elective III	3	0	0	0	6	9
3		Professional Elective IV	3	0	0	0	6	9
4		**Total Credits :**						**43**

** Project (ME7990+) grade assigned in 4th semester*

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	46	49	20	43	40	198

* Grade assigned in fourth semester

LIST OF ELECTIVES

Electives I and II – EVEN Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7100	Production System Design & Control	3	0	0	0	6	9
2	ME7110	Handling Systems Design & Tooling for Automated Manufacturing and Assembly	3	0	0	0	6	9
3	ME7150	Artificial Intelligence in Manufacturing	3	0	0	0	6	9
4	ME7170	Flexible Manufacturing Systems	3	0	0	0	6	9
5	ME7180	Machine Vision and its Applications	3	0	0	0	6	9
6	HS 7410	Management of Finance, Marketing and Personnel	3	0	0	0	6	9
7	ME7400	Mechatronic Systems	3	0	0	0	6	9
8	ME7410	Control Engineering	3	0	0	0	6	9
9	ME7420	Manufacturing Methods in Precision Engineering	3	0	0	0	6	9
10	ME7470	Industrial Instrumentation	3	0	0	0	6	9
11	ME7480	Optical Instrumentation	3	0	0	0	6	9
12	ME7500	Measurement Systems	3	0	0	0	6	9
13	ME7540	Horological Instruments	3	0	0	0	6	9
14	ME7560	Precision Engineering Elements and Instruments	3	0	0	0	6	9
15	ME6014	Micro Manufacturing Technology	3	0	0	0	6	9
16	ME7020	Robotics and Robot Applications	3	0	0	0	6	9

Electives III and IV – ODD Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	MA6310	Operations Research I	3	0	0	0	6	9
2	ME7160	Computational Methods in Design and Manufacture	3	0	0	0	6	9
3	ME7200	Treatment of Materials	3	0	0	0	6	9
4	ME7210	Machine Tool Dynamics	3	0	0	0	6	9
5	ME7220	Metal Removal Processes	3	0	0	0	6	9
6	ME7300	Friction and Wear in Machinery	3	0	0	0	6	9
7	MT 5610	Metal Forming Processes	3	0	0	0	6	9
8	PH 6470	Applied Optics	3	0	0	0	6	9
9	ME7450	Precision Electric Drives and Actuators	3	0	0	0	6	9
10	ME7460	Inertial Instrumentation	3	0	0	0	6	9
11	ME7490	Design and Packaging of Electronic Equipment	3	0	0	0	6	9
12	ME7510	Precision Engineering Design	3	0	0	0	6	9
13	ME7520	Proportional and Servo Hydraulic Controls	3	0	0	0	6	9
14	ME7530	Design of Robotic Manipulators	3	0	0	0	6	9
15	ME7550	Control System Design: A Computer Aided Approach	3	0	0	0	6	9

M.Tech 2017 Batch

Nomenclature

L: Number of Lectures
T: Number of Tutorials
P: Number of Laboratory Sessions
C: Number of Credit

Branch Code: ME1

M.Tech. in MECHANICAL ENGINEERING

STREAM: Thermal Engineering

2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5101	Advanced heat and mass transfer	3	1	0	0	6	10
2	ME5103	Incompressible fluid flow	3	0	0	0	6	9
3	ME5105	Applied Thermodynamics	3	0	0	0	6	9
4	ME5107	Numerical methods in thermal engineering	2	0	0	3	5	10
5	ME5109	Measurements in thermal engineering	2	0	0	3	5	10
		Credits for semester 1						48

Semester 2 (Student must register for at least three out of the five electives during 2nd sem)

S.No	Course No	Course Name	L	T	E	P	O	C
1		Professional elective 1	3	0	0	0	6	9
2		Professional elective 2	3	0	0	0	6	9
3		Professional elective 3	3	0	0	0	6	9
4		Professional elective 4	3	0	0	0	6	9
5		Professional free elective	3	0	0	0	6	9
6	ME5180	Thermal engineering lab	0	0	0	3	0	3
		Credits for semester 2						48

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20
		Total Credits						**20**

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	35	35
		Total Credits :						**35**

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	48	48	20	35	40	191

Ratio of elective-credits to total-credits (excluding project credits): 46.8%

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1.	ME5120	Gas Turbine Engineering	3	0	0	0	6	9
2.	ME5124	Heat and Mass Transfer in Biological Systems	3	0	0	0	6	9
3.	ME5127	Introduction to Atmospheric Science	3	0	0	0	6	9
4.	ME6120	Flow and Thermal Instabilities	3	0	0	0	6	9
5.	ME6122	Jetflow & Acoustics	3	0	0	0	6	9
6.	ME6123	Cavitation	3	0	0	0	6	9
7.	ME6124	Turbomachinery Noise and Control	3	0	0	0	6	9
8.	ME6125	Theory of steam and gas turbines	3	0	0	0	6	9
9.	ME6126	Design of pumps	3	0	0	0	6	9
10.	ME6127	Energy and Environment	3	0	0	0	6	9
11.	ME6128	Aerodynamic Design of Axial Compressors and Turbines	3	0	0	0	6	9
12.	ME6130	Theory and design of centrifugal m/c	3	0	0	0	6	9
13.	ME6132	Theory of axial compressors	3	0	0	0	6	9
14.	ME6134	Airconditioning and ventilation	3	0	0	0	6	9
15.	ME6136	Refrigeration machinery and components	3	0	0	0	6	9
16.	ME6138	Sorption refrigeration and heating systems	3	0	0	0	6	9
17.	ME6139	Simulation of IC Engines Processes	3	0	0	0	6	9
18.	ME6140	HVAC Systems and Applications	3	0	0	0	6	9
19.	ME6141	CFD and Its Applications to Engine Processes	3	0	0	0	6	9
20.	ME6142	Advanced Cryogenic Systems	3	0	0	0	6	9
21.	ME6143	Engine Instrumentation & Electronic Management	3	0	0	0	6	9
22.	ME6144	Thermal Energy Conservation	3	0	0	0	6	9
23.	ME6145	Transport Processes in Engines	3	0	0	0	6	9
24.	ME6146	Utilization of Solar Energy	3	0	0	0	6	9
25.	ME6147	Rocket technology	3	0	0	0	6	9

S.No	Course No	Course Name	L	T	E	P	O	C
26.	ME6148	Renewable Energy Technology	3	0	0	0	6	9
27.	ME6149	Fundamentals of Combustion	3	0	0	0	6	9
28.	ME6154	Design and Optimization of Energy Systems	3	0	0	0	6	9
29.	ME6156	Design of combustion engines	3	0	0	0	6	9
30.	ME6158	Engine systems and performance	3	0	0	0	6	9
31.	ME6160	Alternative Fuels	3	0	0	0	6	9
32.	ME6164	Laser Diagnostics in Engines	3	0	0	0	6	9
33.	ME6166	Combustion technology	3	0	0	0	6	9
34.	ME6168	Air-breathing engines	3	0	0	0	6	9
35.	ME6170	Theory of Fire Propagation	3	0	0	0	6	9
36.	ME7120	Microscale Fluid Flow and Machinery	3	0	0	0	6	9
37.	ME7121	Heat and Mass Transfer in Porous Media	3	0	0	0	6	9
38.	ME8120	Computational Fluid Dynamics of Turbomachinery	3	0	0	0	6	9

Branch Code: ME2

M.Tech. in MECHANICAL ENGINEERING

STREAM: MECHANICAL DESIGN

2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5201	Computational methods in engineering	3	1	0	0	6	10
2	ME5203	Advanced mechanics of solids	3	0	0	0	6	9
3	ME5205	Theory of vibration	3	0	0	0	6	9
4	ME5207	Design with advanced materials	3	0	0	0	6	9
5		Professional elective 1	3	0	0	0	6	9
6	ME5281	Mechanical design lab	0	0	0	3	0	3
		Credits for semester 1						**49**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5204	Finite element analysis	3	0	0	0	6	9
2		Professional elective 2	3	0	0	0	6	9
3		Professional elective 3	3	0	0	0	6	9
4		Professional elective 4	3	0	0	0	6	9
5		Professional free elective	3	0	0	0	6	9
6	ME5280	Design practice using CAD tools	1	0	0	3	2	6
		Credits for semester 2						**51**

Note: (1) Student must register for at least three out of the four electives during the second semester
(2) Student can opt to do professional elective 1 in the third semester

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6490*	Project Phase I	0	0	0	0	20	20
		Total Credits						**20**

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6490	Project Phase II	0	0	0	0	35	35
		Total Credits :						**35**

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6490#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	1	2	Summer	3	4	Total
Credits	49	51	20	35	40	195

Ratio of elective-credits to total-credits (excluding project credits): 45%

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1	CT7000	Composite Materials Science	3	0	0	0	6	9
2	CT7120	Modeling and Equipment Des for Comp Proc	3	0	0	0	6	9
3	ID5020	Multi-body Dynamics and Applications	3	0	0	0	6	9
4	ID6010	Constitutive Modeling in Continuum Mechanics	3	0	0	0	6	9
5	ID7010	Advanced Finite Element Analysis	3	0	0	0	6	9
6	ME6750	Gear Design	3	0	0	0	6	9
7	ME6760	Design of Mechanical Transmission Elements	3	0	0	0	6	9
8	ME6770	Design of Pressure vessels and Piping	3	0	0	0	6	9
9	ME6780	Design Synthesis	3	0	0	0	6	9
10	ME6810	Transmission Mechanisms and Manipulators	3	0	0	0	6	9
11	ME6840	Design for Manufacture and Assembly	3	0	0	0	6	9
12	ME6850	Product Reliability	3	0	0	0	6	9
13	ME6870	CAD/CAM for product design	3	0	0	0	6	9
14	ME7020	Robotics and Robot applications	3	0	0	0	6	9
15	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9
16	ME7300	Friction and Wear in Machinery	3	0	0	0	6	9
17	ME7400	Mechatronic Systems	3	0	0	0	6	9
18	ME7430	Oil Hydraulics and Pneumatic Systems	3	0	0	0	6	9
19	ME7470	Industrial Instrumentation	3	0	0	0	6	9
20	ME7500	Measurement Systems	3	0	0	0	6	9
21	ME7640	Tribo Design and Analysis	3	0	0	0	6	9
22	ME7660	Nonlinear Solid Mechanics	3	0	0	0	6	9
23	ME7680	Optimization Methods for Mechanical Design	3	0	0	0	6	9
24	ME7710	Advanced Vibration and Acoustics	3	0	0	0	6	9
25	ME7740	Structural Health and Integrity Monitoring	3	0	0	0	6	9
26	ME7820	Rotor Dynamics	3	0	0	0	6	9
27	ME7830	Random Vibrations	3	0	0	0	6	9
28	ME7840	Signal Processing in Mechanical systems	3	0	0	0	6	9
29	ME7850	Modal Analysis of Mechanical systems	3	0	0	0	6	9
30	ME7860	Tribo-Instrumentation	3	0	0	0	6	9
31	ME7870	Diagnostic Maintenance	3	0	0	0	6	9
32	ME7880	Vehicular Vibration	3	0	0	0	6	9
33	ME7890	Advanced Applied Finite Element	3	0	0	0	6	9
34	ME7910	Acoustics and Noise Control	3	0	0	0	6	9
35	ME7920	Applied Finite Element	3	0	0	0	6	9
36	ME7930	Chaotic Vibrations	3	0	0	0	6	9
37	ME 7190	Introduction to Fracture Mechanics	3	0	0	0	6	9
38	ME6012	Mechanics of Human Movement	3	0	0	0	6	9
39	ME6710	Theory of Mechanisms	3	0	0	0	6	9
40	ME6720	Failure Analysis and Design	3	0	0	0	6	9
41	ME6016	Mechanics of Thin Films for Microsystem Design	3	0	0	0	6	9
42	ME6003	Variational Principles in Mechanics	3	0	0	0	6	9
43	ME6820	Fundamentals of Engineering Design	3	0	0	0	6	9
44	ME6015	Elastic waves and ultrasonics	3	0	0	0	6	9
45	ID6070	Mechanics of Viscoelastic materials	3	0	0	0	6	9
46	ME8001	Mechanics of Mixtures	3	0	0	0	6	9
47	ME7023	Foundations of computational materials modelling	3	0	0	0	6	9

Any other elective with the approval of the Faculty Advisor/Guide

Branch Code: ME3

M.Tech. in MECHANICAL ENGINEERING
STREAM: MANUFACTURING ENGINEERING
2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5301	Advanced materials processing and characterization	3	0	0	0	6	9
2	ME5303	Computer aided design in manufacturing	3	0	0	0	6	9
3	ME5305	Computer numerical control and adaptive control	3	0	0	0	6	9
4	ME5201	Computational methods in engineering	3	1	0	0	6	10
5		Professional elective 1	3	0	0	0	6	9
	ME5381	Basic manufacturing lab	0	0	0	4	0	4
		Credits for semester 1						**50**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5300	Metrology and computer aided design	3	0	0	0	6	9
2	ME5302	Sensors for intelligent manufacturing and condition monitoring	3	0	0	0	6	9
3		Professional elective 2	3	0	0	0	6	9
4		Professional elective 3	3	0	0	0	6	9
5		Professional elective 4	3	0	0	0	6	9
6	ME5380	Manufacturing and precession engineering lab	0	0	0	4	0	4
		Credits for semester 2						**49**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20
		Total Credits						**20**

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	35	35
		Total Credits :						**35**

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	50	49	20	35	40	194

Ratio of elective-credits to total-credits (excluding project credits): 36.4%

LIST OF ELECTIVES

Electives I and II – EVEN Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6320	Mechatronic Systems	3	0	0	0	6	9
2	ME6320	Micro Manufacturing Technology	3	0	0	0	6	9
3	ME6321	Robotics and Robot Applications	3	0	0	0	6	9
4	ME6323	Production System Design & Control	3	0	0	0	6	9
5	ME6324	Artificial Intelligence in Manufacturing	3	0	0	0	6	9
6	ME6326	Machine Vision and its Applications	3	0	0	0	6	9
7	ME6329	Flexible Manufacturing Systems	3	0	0	0	6	9
8	ME6332	Management of Finance, Marketing and Personnel	3	0	0	0	6	9
9	ME7420	Manufacturing Methods in Precision Engineering	3	0	0	0	6	9

Electives III and IV – ODD Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	MA6310	Operations Research I	3	0	0	0	6	9
2	ME6331	Treatment of Materials	3	0	0	0	6	9
3	ME6333	Metal Removal Processes	3	0	0	0	6	9
4	MM5610	Metal Forming Processes	3	0	0	0	6	9
5	PH6470	Applied Optics	3	0	0	0	6	9

List of courses offered by the department

The syllabus for each of the courses can be found at the academic section website (https://academic.iitm.ac.in/coursecontent) by providing the course number mentioned in the following list. The description of the symbols for the items in the columns 4-9 is given below.

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

The new course numbers in the format MEABXY means:

- **A**:
  - A $<$ 5: Undergraduate level course
  - 5 $\leq$ A $\leq$ 6: Graduate level course
  - A $>$ 6: Advanced graduate level course
- **B**:
  - B=1: Thermal Stream
  - B=2: Design Stream
  - B=3: Manufacturing Stream
- **XY**: A unique two-digit course number
  - Odd Number: Offered in odd semester
  - Even Number: Offered in even semester

Curriculum Tables

Department of Mechanical Engineering

Indian Institute of Technology Madras

Last updated: 17-November-2017

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Nomenclature

Students joined before 2015

L: Number of Lectures
T: Number of Tutorials
P: Number of Laboratory Sessions
C: Number of Credit

SCY: Basic Sciences-Chemistry
SPH: Basic Sciences-Physics
SMA: Basic Sciences-Mathematics
BET: Basic Engineering Theory
BES: Basic Engineering Practice
PMT: Professional Major Theory
HPF: Humanities Professional Compulsory
SLS: Life Sciences
PML: Professional Major Laboratory
HSS: Humanities and Scoial Sciences Elective
MNS: Minor Stream Elective
PIT: Professional Industrial Training
PIL: Professional Industrial Lecture
PSS: Professional Self-Study Course
PMP: Professional Major Project Work

Students joined in 2015 and later

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

S: Basic Science
E: Basic Engineering
P: Professional Engineering
H: Humanities

B.Tech Curriculum

B.Tech 2014 Batch

ME - B.Tech. Mechanical Engineering
2014-Batch

SEMESTER 1

S.No	Course No	Course Name	L	T	P	C	Category
1	CY1002	Chemistry Lab I	1	0	3	3	SCY
2	ID1100	Concepts in Engineering Design	2	0	0	2	BET
3	MA1010	Calculus I Functions of One Variable	3	1	0	4	SMA
4	ME1100	Thermodynamics	3	0	0	3	BET
5	ME1110	Introduction to Mechanical Engineering	2	0	0	2	PMT
6	ME1120	Engineering Drawing	3	0	0	3	BES
7	PH1010	Physics I	3	0	0	3	SPH
8	PH1030	Physics Laboratory I	0	0	3	2	SPH
9	GN1100	Life Skills	2	0	0	0	HPF
		Total Credits :				**24**	

SEMESTER 2

S.No	Course No	Course Name	L	T	P	C	Category
1	AM1100	Engineering Mechanics	4	0	0	4	BET
2	CS1100	Computational Engineering	3	0	0	3	BET
3	CY1001	Chemistry: Structure, Bonding & Reactivity	4	0	0	4	SCY
4	MA1020	Calculus II Functions of Several Variables	3	1	0	4	SMA
5	PH1020	Physics II	3	0	0	3	SPH
6	PH1040	Physics Laboratory II				2	SPH
		Total Credits :				**20**	
	WS1010	Workshop I	0	0	6	4	BES
	WS1020	Workshop II	0	0	3	2	BES

SEMESTER 3

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2200	Strength of Materials	4	0	0	4	PMT
2	BT1010	Life Sciences	2	0	0	2	SLS
3	EE1100	Basic Electrical Engineering	3	0	0	3	BET
4	ID1200	Ecology and Environment	2	0	0	2	BET
5	ME2050	Machine Drawing Practice	1	0	6	5	PML
6	HSE1	Humanities Elective 1	3	0	0	3	HSS
7	MAE1	Mathematics Elective 1	3	0	0	3	SMA
		Total Credits :				**22**	

SEMESTER 4

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2530	Foundations of Fluid Mechanics	3	1	0	4	PMT
2	AM2540	Applied Mechanics Lab	0	0	3	2	PML
3	EE2100	Electrical Sciences	3	0	0	3	PMT
4	ME2220	Kinematics and Dynamics of Machinery	4	0	0	4	PMT
5	ME2240	Instrumentation and Control	4	0	0	4	PMT
6	ME2260	Materials and Design	4	0	0	4	PMT
7	ME2280	Manufacturing Technology	4	0	0	4	PMT
8	MAE4	Mathematics Elective 4	3	0	0	3	SMA
		Total Credits :				**28**	

SEMESTER 5

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3170	Heat Transfer	4	0	0	4	PMT
2	ME3190	Machine Tools & Metrology	4	0	0	4	PMT
3	ME3270	Mechanical Engg Lab I	0	0	3	2	PML
4	ME3310	Turbomachines	4	0	0	4	PMT
5	ME3330	Internal Combustion Engines	3	0	0	3	PMT
6	ME3350	Design of Machine Elements	4	0	0	4	PMT
7	MNS1	Minor Elective 1	3	0	0	3	MNS
		Total Credits :				**24**	

SEMESTER 6

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3250	Refrigeration & Airconditioning	3	0	0	3	PMT
2	ME3260	Thermal Power Engineering	4	0	0	4	PMT
3	ME3280	Mechanical Engineering Lab II	0	0	3	2	PML
4	ME3300	Mechanical Engg. Lab III	0	0	3	2	PML
5	ME3500	Summer Training	0	0	0	2	PIT
6	DPE1	Department Elective 1	3	0	0	3	PMT
7	FRE1	Free Elective 1	3	0	0	3	
8	HSE2	Humanities Elective 2	3	0	0	3	HSS
9	MNS2	Minor Elective 2	3	0	0	3	MNS
		Total Credits :				**25**	

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	IL4020	Industrial Lecture	0	0	0	1	PIL
3	ME4545	Viva voce	0	0	0	2	PMP
4	ME4550	Project I	0	0	0	3	PMP
5	DPE2	Department Elective 2	3	0	0	3	PMT
6	DPE3	Department Elective 3	3	0	0	3	PMT
7	FRE2	Free Elective 2	3	0	0	3	
8	MNS3	Minor Elective 3	3	0	0	3	MNS
9	SSE1	Self-Study Elective 1	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	ME4560	Project II	0	0	0	6	PMP
2	DPE4	Department Elective 4	3	0	0	3	PMT
3	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**12**	

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Req. Cr.	24	20+6	22	28	24	25	23	12	184

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	13	11

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	67	0	3	7	14	10	2	6	0

Remarks

1. Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

2. Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

3. **B.Tech. project**
   - At the end of 6th semester students may opt to do project work or course work
   - Students taking the project option should have satisfactory performance in the mini project in order to continue the major project
   - In lieu of projects, student can do courses for 9 credits (3 credits in 7th semester & 6 credits In 8th semester)
   - Students who fail in mini project shall repeat it or substitute with a course having equivalent credit

4. **B.Tech. Honours**: (Total credit requirement: 184+12 = 196)

   Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.

   1. Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
   2. Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
   3. Project work is a must for B.Tech (Honours) students in B.Tech programme.
   4. Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
   5. Extra credits earned by the students will be counted for prizes and awards.

B.Tech 2015 Batch

ME - B.Tech. in Mechanical Engineering
2015 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1100	Life Skills	0	0	0	0	3	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop II	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2220	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2260	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3170	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 2	3	0	0	0	6	9	P
2		Professional elective 3	3	0	0	0	6	9	P
3		Professional elective 4/ Free elective 4	3	0	0	0	6	9	P
4		Professional elective 5/ Free elective 5	3	0	0	0	6	9	P
5		Professional elective 6/ Free elective 6	3	0	0	0	6	9	P
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	0	0	0	0	2	0	H
8		**Credits for semester 7**						**54**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free or professional elective 4 or project*	3	0	0	0	6	9	
2		Free or professional elective 5 or project*	3	0	0	0	6	9	
3		Free or professional elective 6 or project*	3	0	0	0	6	9	
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
6		Ecology and environment	0	0	0	0	2	0	S
		Credits for semester 8						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Note:

(1) Project in other department can be taken only in lieu of Free Electives

(2) Project in ME Department can be taken in lieu of free or departmental electives

(3) It may be noted that any combination of free electives should lead to 72 credits and not

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Credits	49	56+6	52	59	59	52	54	45	432

***Please note that the indicated credits are only for core program.**

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	177+27	27	84	72	432

B.Tech. Curriculum (from July 2015)

Category	Minimum Credit Requirements (CTF)/BOG	ME
Basic Science (S)	84	84
Basic Eng. (E)	45	45
Profession (P)	180	204
Humanities (H)	27	27
Unallocated credits	96	72
Total	**432**	**432**

B.Tech. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	54	204
Humanities (H)	27	27
Unallocated credits	72	72
Total	**162**	**432**
Percentage	**37.5%**	

BTech (honours): (Total credit requirement: 432 + 27 = **459)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

B.Tech 2016 Batch

ME - B.Tech. in Mechanical Engineering
2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1101	Life Skills I	0	0	0	0	2	0	
8	ID1200	Ecology and Environment	2	0	0	0	0	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop II	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2220	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 2	3	0	0	0	6	9	P
2		Professional elective 3	3	0	0	0	6	9	P
3		Professional elective 4/ Free elective 4	3	0	0	0	6	9	P
4		Professional elective 5/ Free elective 5	3	0	0	0	6	9	P
5		Professional elective 6/ Free elective 6	3	0	0	0	6	9	P
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	0	0	0	0	2	0	H
8		**Credits for semester 7**						**54**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free or professional elective 4 or project*	3	0	0	0	6	9	
2		Free or professional elective 5 or project*	3	0	0	0	6	9	
3		Free or professional elective 6 or project*	3	0	0	0	6	9	
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
6		Ecology and environment	0	0	0	0	2	0	S
		Credits for semester 8						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Note:

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Humanities (H)	27	27
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B.Tech. - Category wise credit Requirements

Category	ME	
	Electives	Total
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- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical

B.Tech 2017 Batch

ME - B.Tech. in Mechanical Engineering
2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1101	Life Skills I	0	0	0	0	2	0	
8	ID1200	Ecology and Environment	2	0	0	0	0	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1301	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1302	Workshop II	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 2	3	0	0	0	6	9	P
2		Professional elective 3	3	0	0	0	6	9	P
3		Professional elective 4/ Free elective 4	3	0	0	0	6	9	P
4		Professional elective 5/ Free elective 5	3	0	0	0	6	9	P
5		Professional elective 6/ Free elective 6	3	0	0	0	6	9	P
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	0	0	0	0	2	0	H
8		**Credits for semester 7**						**54**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free or professional elective 4 or project*	3	0	0	0	6	9	
2		Free or professional elective 5 or project*	3	0	0	0	6	9	
3		Free or professional elective 6 or project*	3	0	0	0	6	9	
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
6		Ecology and environment	0	0	0	0	2	0	S
		Credits for semester 8						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Note:

(1) Project in other department can be taken only in lieu of Free Electives
(2) Project in ME Department can be taken in lieu of free or departmental electives
(3) It may be noted that any combination of free electives should lead to 72 credits and not necessarily 8 courses

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Credits	49	56+6	52	59	59	52	54	45	432

***Please note that the indicated credits are only for core program.**

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	177+27	27	84	72	432

B.Tech. Curriculum (from July 2015)

Category	Minimum Credit Requirements (CTF)/BOG	ME
Basic Science (S)	84	84
Basic Eng. (E)	45	45
Profession (P)	180	204
Humanities (H)	27	27
Unallocated credits	96	72
Total	**432**	**432**

B.Tech. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	54	204
Humanities (H)	27	27
Unallocated credits	72	72
Total	**162**	**432**
Percentage	**37.5%**	

BTech (honours): (Total credit requirement: 432 + 27 = **459)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical

B.Tech Dual Degree Curriculum

B.Tech DD 2013 Batch

BRANCH CODE: ME25

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

2013-Batch

SEMESTER 1

S.No	Course No	Course Name	L	T	P	C	Category
1	CY1002	Chemistry Lab I	1	0	3	3	SCY
2	ID1100	Concepts in Engineering Design	2	0	0	2	BET
3	MA1010	Calculus I Functions of One Variable	3	1	0	4	SMA
4	ME1100	Thermodynamics	3	0	0	3	BET
5	ME1110	Introduction to Mechanical Engineering	2	0	0	2	PMT
6	ME1120	Engineering Drawing	3	0	0	3	BES
7	PH1010	Physics I	3	0	0	3	SPH
8	PH1030	Physics Laboratory I	0	0	3	2	SPH
9	GN1100	Life Skills	2	0	0	2	HPF
		Total Credits :				**24**	

SEMESTER 2

S.No	Course No	Course Name	L	T	P	C	Category
1	AM1100	Engineering Mechanics	4	0	0	4	BET
2	CS1100	Computational Engineering	3	0	0	3	BET
3	CY1001	Chemistry: Structure, Bonding & Reactivity	4	0	0	4	SCY
4	MA1020	Calculus II Functions of Several Variables	3	1	0	4	SMA
5	PH1020	Physics II	3	0	0	3	SPH
6	PH1040	Physics Laboratory II				2	SPH
		Total Credits :				**20**	
	WS1010	Workshop I	0	0	6	4	BES
	WS1020	Workshop II	0	0	3	2	BES

SEMESTER 3

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2200	Strength of Materials	4	0	0	4	PMT
2	BT1010	Life Sciences	2	0	0	2	SLS
3	EE1100	Basic Electrical Engineering	3	0	0	3	BET
4	ID1200	Ecology and Environment	2	0	0	2	BET
5	ME2050	Machine Drawing Practice	1	0	6	5	PML
6	HSE1	Humanities Elective 1	3	0	0	3	HSS
7	MAE1	Mathematics Elective 1	3	0	0	3	SMA
		Total Credits :				**22**	

SEMESTER 4

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2530	Foundations of Fluid Mechanics	3	1	0	4	PMT
2	AM2540	Applied Mechanics Lab	0	0	3	2	PML
3	EN2100	Electrical Sciences	3	0	0	3	PMT
4	ME2220	Kinematics and Dynamics of Machinery	4	0	0	4	PMT
5	ME2240	Instrumentation and Control	4	0	0	4	PMT
6	ME2260	Materials and Design	4	0	0	4	PMT
7	ME2280	Manufacturing Technology	4	0	0	4	PMT
8	MAE4	Mathematics Elective 4	3	0	0	3	SMA
		Total Credits :				**28**	

SEMESTER 5

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3170	Heat Transfer	4	0	0	4	PMT
2	ME3190	Machine Tools & Metrology	4	0	0	4	PMT
3	ME3270	Mechanical Engg Lab I	0	0	3	2	PML
4	ME3310	Turbomachines	4	0	0	4	PMT
5	ME3330	Internal Combustion Engines	3	0	0	3	PMT
6	ME3350	Design of Machine Elements	4	0	0	4	PMT
7	MNS1	Minor Elective 1	3	0	0	3	MNS
		Total Credits :				**24**	

SEMESTER 6

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3250	Refrigeration & Airconditioning	3	0	0	3	PMT
2	ME3260	Thermal Power Engineering	4	0	0	4	PMT
3	ME3280	Mechanical Engineering Lab II	0	0	3	2	PML
4	ME3300	Mechanical Engg. Lab III	0	0	3	2	PML
5	ME3500	Summer Training	0	0	0	2	PIT
6	DPE1	Department Elective 1	3	0	0	3	PMT
7	FRE1	Free Elective 1	3	0	0	3	
8	HSE2	Humanities Elective 2	3	0	0	3	HSS
9	MNS2	Minor Elective 2	3	0	0	3	MNS
		Total Credits :				**25**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Product Design

2013-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6800	Finite Element Analysis	3	0	0	3	PMT
3	ME6820	Fundamentals of Engineering Design	3	0	0	3	PMT
4	ME6830	Principles of Product Design	3	0	0	3	PMT
5	ME6840	Design for Manufacture and Assembly	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6780	Design Synthesis	3	0	0	3	PMT
3	ME6850	Product Reliability	3	0	0	3	PMT
4	ME6870	CAD/CAM for Product Design	3	0	0	3	PMT
5	ME6880	Product Design Lab	0	0	3	2	PML
6	ME7400	Mechatronic Systems	3	0	0	3	PMT
7	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
8	HSE4	Humanities Elective 4	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
3	DDE2	Dual Degree Elective 3	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.
- Any two PMT electives can be taken as Pass/Fail courses
- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.
- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Intelligent Manufacturing
2013-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME7010	Microprocessors in Automation	3	0	0	3	PMT
3	ME7020	Robotics and Robot Applications	3	0	0	3	PMT
4	ME7050	CNC and Adaptive Control	3	0	0	3	PMT
5	ME7240	Modelling and Simulation in Manufacturing	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective 1	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME7120	Sensors for Intelligent Manufacturing	3	0	0	3	PMT
3	ME7130	Computer Int. Manuf. Lab I	0	0	3	2	PML
4	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
5	ME7250	Rapid Response Manufacturing	3	0	0	3	PMT
6	ME7400	Mechatronic Systems	3	0	0	3	PMT
7	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
3	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.
- Any two PMT electives can be taken as Pass/Fail courses
- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.
- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Thermal Engineering

2013-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6041	Incompressible Fluid Flow	3	0	0	3	PMT
3	ME6140	Applied Thermodynamics	3	0	0	3	PMT
4	ME6280	Design & Optimisation of Energy Systems	3	0	0	3	PMT
5	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6080	Measurement in Thermal Engg	4	0	0	4	PMT
3	ME6150	Numerical Methods in Thermal Engg	4	0	0	4	PMT
4	ME6290	Advanced Energy Conversion	3	0	0	3	PMT
6	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
7	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
3	DDE5	Dual Degree Elective 5	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	13	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

- Any two PMT electives can be taken as Pass/Fail courses

- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

B.Tech DD 2014 Batch

BRANCH CODE: ME25

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

2014-Batch

SEMESTER 1

S.No	Course No	Course Name	L	T	P	C	Category
1	CY1002	Chemistry Lab I	1	0	3	3	SCY
2	ID1100	Concepts in Engineering Design	2	0	0	2	BET
3	MA1010	Calculus I Functions of One Variable	3	1	0	4	SMA
4	ME1100	Thermodynamics	3	0	0	3	BET
5	ME1110	Introduction to Mechanical Engineering	2	0	0	2	PMT
6	ME1120	Engineering Drawing	3	0	0	3	BES
7	PH1010	Physics I	3	0	0	3	SPH
8	PH1030	Physics Laboratory I	0	0	3	2	SPH
9	GN1100	Life Skills	2	0	0	2	HPF
		Total Credits :				**24**	

SEMESTER 2

S.No	Course No	Course Name	L	T	P	C	Category
1	AM1100	Engineering Mechanics	4	0	0	4	BET
2	CS1100	Computational Engineering	3	0	0	3	BET
3	CY1001	Chemistry: Structure, Bonding & Reactivity	4	0	0	4	SCY
4	MA1020	Calculus II Functions of Several Variables	3	1	0	4	SMA
5	PH1020	Physics II	3	0	0	3	SPH
6	PH1040	Physics Laboratory II				2	SPH
		Total Credits :				**20**	
	WS1010	Workshop I	0	0	6	4	BES
	WS1020	Workshop II	0	0	3	2	BES

SEMESTER 3

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2200	Strength of Materials	4	0	0	4	PMT
2	BT1010	Life Sciences	2	0	0	2	SLS
3	EE1100	Basic Electrical Engineering	3	0	0	3	BET
4	ID1200	Ecology and Environment	2	0	0	2	BET
5	ME2050	Machine Drawing Practice	1	0	6	5	PML
6	HSE1	Humanities Elective 1	3	0	0	3	HSS
7	MAE1	Mathematics Elective 1	3	0	0	3	SMA
		Total Credits :				**22**	

SEMESTER 4

S.No	Course No	Course Name	L	T	P	C	Category
1	AM2530	Foundations of Fluid Mechanics	3	1	0	4	PMT
2	AM2540	Applied Mechanics Lab	0	0	3	2	PML
3	EN2100	Electrical Sciences	3	0	0	3	PMT
4	ME2220	Kinematics and Dynamics of Machinery	4	0	0	4	PMT
5	ME2240	Instrumentation and Control	4	0	0	4	PMT
6	ME2260	Materials and Design	4	0	0	4	PMT
7	ME2280	Manufacturing Technology	4	0	0	4	PMT
8	MAE4	Mathematics Elective 4	3	0	0	3	SMA
		Total Credits :				**28**	

SEMESTER 5

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3170	Heat Transfer	4	0	0	4	PMT
2	ME3190	Machine Tools & Metrology	4	0	0	4	PMT
3	ME3270	Mechanical Engg Lab I	0	0	3	2	PML
4	ME3310	Turbomachines	4	0	0	4	PMT
5	ME3330	Internal Combustion Engines	3	0	0	3	PMT
6	ME3350	Design of Machine Elements	4	0	0	4	PMT
7	MNS1	Minor Elective 1	3	0	0	3	MNS
		Total Credits :				**24**	

SEMESTER 6

S.No	Course No	Course Name	L	T	P	C	Category
1	ME3250	Refrigeration & Airconditioning	3	0	0	3	PMT
2	ME3260	Thermal Power Engineering	4	0	0	4	PMT
3	ME3280	Mechanical Engineering Lab II	0	0	3	2	PML
4	ME3300	Mechanical Engg. Lab III	0	0	3	2	PML
5	ME3500	Summer Training	0	0	0	2	PIT
6	DPE1	Department Elective 1	3	0	0	3	PMT
7	FRE1	Free Elective 1	3	0	0	3	
8	HSE2	Humanities Elective 2	3	0	0	3	HSS
9	MNS2	Minor Elective 2	3	0	0	3	MNS
		Total Credits :				**25**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Product Design

2014-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6800	Finite Element Analysis	3	0	0	3	PMT
3	ME6820	Fundamentals of Engineering Design	3	0	0	3	PMT
4	ME6830	Principles of Product Design	3	0	0	3	PMT
5	ME6840	Design for Manufacture and Assembly	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6224	Design Synthesis	3	0	0	3	PMT
3	ME6226	Product Reliability	3	0	0	3	PMT
4	ME6228	CAD/CAM for Product Design	3	0	0	3	PMT
5	ME6880	Product Design Lab	0	0	3	2	PML
6	ME6332	Mechatronic Systems	3	0	0	3	PMT
7	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
8	HSE4	Humanities Elective 4	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
3	DDE2	Dual Degree Elective 3	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.
- Any two PMT electives can be taken as Pass/Fail courses
- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.
- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Intelligent Manufacturing
2014-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME7010	Microprocessors in Automation	3	0	0	3	PMT
3	ME7020	Robotics and Robot Applications	3	0	0	3	PMT
4	ME7050	CNC and Adaptive Control	3	0	0	3	PMT
5	ME5307	Modelling and Simulation in Manufacturing	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective 1	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME5302	Sensors for Intelligent Manufacturing	3	0	0	3	PMT
3	ME7130	Computer Int. Manuf. Lab I	0	0	3	2	PML
4	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
5	ME7250	Rapid Response Manufacturing	3	0	0	3	PMT
6	ME7400	Mechatronic Systems	3	0	0	3	PMT
7	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
3	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	15	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT
Req. Cr.	85	6	3	7	14	10	2	6	0

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.
- Any two PMT electives can be taken as Pass/Fail courses
- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.
- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 52 & 53)

STREAM: Thermal Engineering

2014-Batch

SEMESTER 7

S.No	Course No	Course Name	L	T	P	C	Category
1	HS3050	Professional Ethics	2	0	0	2	HPF
2	ME6040	Incompressible Fluid Flow	3	0	0	3	PMT
3	ME5105	Applied Thermodynamics	3	0	0	3	PMT
4	ME6280	Design & Optimisation of Energy Systems	3	0	0	3	PMT
5	DDE1	Dual Degree Elective 1	3	0	0	3	PMT
6	FRE2	Free Elective 2	3	0	0	3	
7	MNS3	Minor Elective 3	3	0	0	3	MNS
8	SSE1	Self-Study Elective	3	0	0	3	PSS
		Total Credits :				**23**	

SEMESTER 8

S.No	Course No	Course Name	L	T	P	C	Category
1	IL4040	Industrial Lecture	0	0	0	1	PIL
2	ME6080	Measurement in Thermal Engg	4	0	0	4	PMT
3	ME5107	Numerical Methods in Thermal Engg	4	0	0	4	PMT
4	ME6150	Advanced Energy Conversion	3	0	0	3	PMT
6	DDE2	Dual Degree Elective 2	3	0	0	3	PMT
7	DDE3	Dual Degree Elective 3	3	0	0	3	PMT
8	HSE3	Humanities Elective 3	3	0	0	3	HSS
		Total Credits :				**21**	

SEMESTER 9

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980*	Project I - DD	0	0	0	6*	PMP
2	DDE4	Dual Degree Elective 4	3	0	0	3	PMT
3	DDE5	Dual Degree Elective 5	3	0	0	3	PMT
		Total Credits :				**6**	

Project (ME7980*) grade will be awarded in 10th semester

SEMESTER 10

S.No	Course No	Course Name	L	T	P	C	Category
1	ME7980	Project II	0	0	0	15	PMP
		Total Credits :				**15**	

Semester	I	II	III	IV	V	VI	VII	VIII	IX	X	Total
Req. Cr.	24	20+6	22	28	24	25	23	21	6	15	214

Category	BES	BET	HPF	HSS	MNS	PIL	PIT	PML	PMP
Req. Cr.	9	17	4	9	9	1	2	13	15

Category	PMT	PPF	PSS	SCY	SMA	SPH	SLS	FRE	EXT

Remarks

- Electives I to IV from Mechanical Elective List or other equivalents from other departments with the consent of the Faculty Adviser.

- Any two PMT electives can be taken as Pass/Fail courses

- Free Electives I & II category will be indicated in the grade card according to the course chosen by the student from own department or from other departments such as MA, HS & PH.

- Dual Degree (B.Tech. Honours & M.Tech.) - (Total credit requirement: 214+12 = 226)
  - Students having a CGPA of ≥ 8.5 and who have cleared all the courses in the first attempt can opt for B.Tech (Honours) at the end of the fourth semester.
  - Those who opt for B.Tech (Honours) must register for 12 additional PMT credits.
  - Students can register for a maximum of 2 additional PMT courses in a semester from the Honours basket of courses over and above the courses prescribed in the B.Tech curriculum from 5th semester onwards.
  - Project work is a must for B.Tech (Honours) students in B.Tech programme.
  - Those who have registered for B.Tech (Honours) must maintain an average CGPA of 8.5 from 4th semester onwards. Otherwise, they will be awarded only B.Tech Degree.
  - Extra credits earned by the students will be counted for prizes and awards.

B.Tech DD 2015 Batch

Branch Code: ME25
Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering
2015-Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	GN1100	Life Skills	0	0	0	0	3	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop I	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honours elective 1	3	0	0	0	6	9	
		Credits for semester 6 for Honours						**61**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Mechanical Design

2015-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional Elective 2	3	0	0	0	6	9	P
2	ME5205	Theory of Vibration	3	0	0	0	6	9	P
3	ME5207	Design with Advanced Materials	3	0	0	0	6	9	P
4		Free Elective 4	3	0	0	0	6	9	
5		Free Elective 5	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	2	0	0	0	0	0	H
8	ME5281	Mechanical Design Laboratory	0	0	0	3	0	3	P
		Credits for semester 7						**57**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**66**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME5203	Advanced Mechanics of Solids	3	0	0	0	6	9	P
2		Professional Elective 3	3	0	0	0	6	9	P
3		Professional Elective 4	3	0	0	0	6	9	P
4		Free Elective 6	3	0	0	0	6	9	
5		Free Elective 7	3	0	0	0	6	9	
6	ME5280	Design Practice with CAD Tools	1	0	0	3	2	6	P
		Credits for semester 8						**51**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**60**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490#	Project phase 1	0	0	0	0	22	22	
		Credits for summer						**22**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490+	Project phase 2	0	0	0	0	23	23	
2	ME	Professional Elective 5	3	0	0	0	6	9	P
3		Free Elective 8	3	0	0	0	6	9	
4	ME5204	Finite Element Analysis	3	0	0	0	6	9	P
		Credits for semester 9						**50**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	57	51	22	50	40	553

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**168+72+85**	**27**	**84**	**72**	**553**

DD Curriculum Mechanical Design

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	240
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**553**

DD Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	240
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**468 + 85**
Percentage		

Note:

- **Professional Elective 1 - to be chosen from 4000 level;**
- **Professional Elective 2-4 - to be chosen from 5000 + level.**

BTech (honours) + M.Tech: (Total credit requirement: 553 + 27 = **580)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No.31 & 32)

STREAM: Intelligent Manufacturing
2015-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7240	Modeling and Simulation in Manufacturing	3	0	0	0	6	9	P
2	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9	P
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9	P
4	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9	P
5		Free Elective 4	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	0	0	0	0	2	0	H
		Credits for semester 7						**54**	
	ME	**Honours elective 2**	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free Elective 5	3	0	0	0	6	9	
2		Free Elective 6	3	0	0	0	6	9	
3	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9	P
4	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9	P
5	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4	P
6	ME	Professional Elective 2	3	0	0	0	6	9	P
		Credits for semester 7						**49**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**58**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2	ME	Professional Elective 3	3	0	0	0	6	9	P
3	ME	Professional Elective 4	3	0	0	0	6	9	P
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	54	49	21	60	40	557

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	172+72+85	27	84	72	557

DD Curriculum Intelligent Manufacturing

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	244
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	244
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**472 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 557 + 27 = **584)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Thermal Engineering
2015-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6080	Measurement in Thermal Engg	2	0	0	3	5	10	P
2	ME6150	Numerical Methods in Thermal Engg	2	0	0	3	5	10	P
3		Professional elective 2	3	0	0	0	6	9	P
4		Free elective 4	3	0	0	0	6	9	
5		Free elective 5	3	0	0	0	6	9	
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	2	0	0	0	2	0	H
8		**Credits for semester 7**						**56**	
		Honours elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**65**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective 3	3	0	0	0	6	9	P
2		Professional elective 4	3	0	0	0	6	9	P
3		Professional elective 5	3	0	0	0	6	9	P
4		Professional elective 6	3	0	0	0	6	9	P
5		Professional elective 7	3	0	0	0	6	9	P
6		**Credits for semester 8**						**45**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**54**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2		Free elective 6	3	0	0	0	6	9	
3		Free elective 7	3	0	0	0	6	9	
4		Free elective 8	3	0	0	0	6	9	
5		Professional elective 8	3	0	0	0	6	9	P
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	56	45	21	60	40	555

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**170+72+85**	**27**	**84**	**72**	**555**

DD Curriculum Thermal stream

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	242
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	242
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**470 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 555 + 27 = **582)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

B.Tech DD 2016 Batch

Branch Code: ME25
Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering
2016-Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8	GN1101	Life Skills I	0	0	0	0	2	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1120	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop I	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Mechanical Design

2016-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional Elective 2	3	0	0	0	6	9	P
2	ME5205	Theory of Vibration	3	0	0	0	6	9	P
3	ME5207	Design with Advanced Materials	3	0	0	0	6	9	P
4		Free Elective 4	3	0	0	0	6	9	
5		Free Elective 5	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	2	0	0	0	0	0	H
8	ME5281	Mechanical Design Laboratory	0	0	0	3	0	3	P
		Credits for semester 7						**57**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**66**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME5203	Advanced Mechanics of Solids	3	0	0	0	6	9	P
2		Professional Elective 3	3	0	0	0	6	9	P
3		Professional Elective 4	3	0	0	0	6	9	P
4		Free Elective 6	3	0	0	0	6	9	
5		Free Elective 7	3	0	0	0	6	9	
6	ME5280	Design Practice with CAD Tools	1	0	0	3	2	6	P
		Credits for semester 8						**51**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**60**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490#	Project phase 1	0	0	0	0	22	22	
		Credits for summer						**22**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490+	Project phase 2	0	0	0	0	23	23	
2	ME	Professional Elective 5	3	0	0	0	6	9	P
3		Free Elective 8	3	0	0	0	6	9	
4	ME5204	Finite Element Analysis	3	0	0	0	6	9	P
		Credits for semester 9						**50**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	57	51	22	50	40	553

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**168+72+85**	**27**	**84**	**72**	**553**

DD Curriculum Mechanical Design

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	240
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**553**

DD Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	240
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**468 + 85**
Percentage		

Note:

- **Professional Elective 1 - to be chosen from 4000 level;**
- **Professional Elective 2-4 - to be chosen from 5000 + level.**

BTech (honours) + M.Tech: (Total credit requirement: 553 + 27 = **580)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No.31 & 32)

STREAM: Intelligent Manufacturing
2016-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7240	Modeling and Simulation in Manufacturing	3	0	0	0	6	9	P
2	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9	P
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9	P
4	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9	P
5		Free Elective 4	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	0	0	0	0	2	0	H
		Credits for semester 7						**54**	
	ME	**Honours elective 2**	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free Elective 5	3	0	0	0	6	9	
2		Free Elective 6	3	0	0	0	6	9	
3	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9	P
4	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9	P
5	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4	P
6	ME	Professional Elective 2	3	0	0	0	6	9	P
		Credits for semester 7						**49**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**58**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2	ME	Professional Elective 3	3	0	0	0	6	9	P
3	ME	Professional Elective 4	3	0	0	0	6	9	P
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	54	49	21	60	40	557

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	172+72+85	27	84	72	557

DD Curriculum Intelligent Manufacturing

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	244
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	244
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**472 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 557 + 27 = **584)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Thermal Engineering
2016-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6080	Measurement in Thermal Engg	2	0	0	3	5	10	P
2	ME6150	Numerical Methods in Thermal Engg	2	0	0	3	5	10	P
3		Professional elective 2	3	0	0	0	6	9	P
4		Free elective 4	3	0	0	0	6	9	
5		Free elective 5	3	0	0	0	6	9	
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	2	0	0	0	2	0	H
8		**Credits for semester 7**						**56**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**65**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective3	3	0	0	0	6	9	P
2		Professional elective 4	3	0	0	0	6	9	P
3		Professional elective 5	3	0	0	0	6	9	P
4		Professional elective 6	3	0	0	0	6	9	P
5		Professional elective 7	3	0	0	0	6	9	P
6		**Credits for semester 8**						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2		Free elective 6	3	0	0	0	6	9	
3		Free elective 7	3	0	0	0	6	9	
4		Free elective 8	3	0	0	0	6	9	
5		Professional elective 8	3	0	0	0	6	9	P
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	56	45	21	60	40	555

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**170+72+85**	**27**	**84**	**72**	**555**

DD Curriculum Thermal stream

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	242
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**555**

DD. - Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	242
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**470 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 555 + 27 = **582)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

B.Tech DD 2017 Batch

Branch Code: ME25
Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering
2017-Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1101	Functions of several variables	3	1	0	0	6	10	S
2	PH1010	Physics 1	3	1	0	0	6	10	S
3	CS1100	Introduction to programming	3	0	0	3	6	12	E
4	ME1100	Thermodynamics	3	1	0	0	6	10	P
5	PH1030	Physics lab	0	0	0	4	0	4	S
6	CY1002	Chemistry lab	0	0	0	3	0	3	S
7	ID1200	Ecology and Environment	2	0	0	0	0	0	
8	GN1101	Life Skills I	0	0	0	0	2	0	
		NCC/ NSS/ NSO	0	0	0	0	2	0	
		Credits for semester 1						**49**	

Winter

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1010	Workshop I	0	0	0	3	0	3	E

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA1102	Series and matrices	3	1	0	0	6	10	S
2	PH1020	Physics 2	3	1	0	0	6	10	S
3	CY1001	Chemistry	3	1	0	0	6	10	S
4	AM1100	Engineering Mechanics	3	1	0	0	6	10	E
5	ME1480	Engineering drawing	1	0	0	3	3	7	E
6	HS	Humanities elective 1	3	0	0	0	6	9	H
7	GN1102	Life Skills II	0	0	0	0	1	0	
8		NCC/ NSS/ NSO	0	0	0	0	3	0	
		Credits for semester 2						**56**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	WS1020	Workshop I	0	0	0	3	0	3	E

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA2020	Differential equations	3	0	0	0	6	9	S
2	EE1100	Basic electrical engineering	3	1	0	0	6	10	E
3	AM2200	Strength of materials	3	1	0	0	6	10	P
4	AM2530	Foundations of fluid mechanics	3	1	0	0	6	10	P
5	ME2201	Kinematics and dynamics of machinery	3	1	0	0	6	10	P
6	AM2540	Applied mechanics/ Fluid mechanics lab	0	0	0	3	0	3	P
		Credits for semester 3						**52**	

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	MA	Mathematics elective	3	0	0	0	6	9	S
2	ME2100	Applied thermal engineering	3	1	0	0	6	10	P
3	ME2200	Materials and design	3	1	0	0	6	10	P
4	ME2300	Manufacturing processes	3	1	0	0	6	10	P
5	ME2400	Measurements and instrumentation	3	0	0	2	6	11	P
6	HS	Humanities elective 2	3	0	0	0	6	9	H
		Credits for semester 4						**59**	

Semester 5

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME3101	Heat transfer	3	1	0	0	6	10	P
2	ME3103	Energy conversion systems	3	1	0	0	6	10	P
3	ME3201	Design of machine elements	3	1	0	0	6	10	P
4	ME3301	Manufacturing technology	3	1	0	0	6	10	P
5		Free elective 1	3	0	0	0	6	9	
6	ME3481	Mechanical engineering lab 1	0	0	0	3	0	3	P
7	ME3281	Machine drawing practice	1	0	0	3	3	7	P
		Credits for semester 5						**59**	

Semester 6

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional elective 1	3	0	0	0	6	9	P
2		Free elective 2	3	0	0	0	6	9	
3		Free elective 3	3	0	0	0	6	9	
4	BT1010	Life sciences	3	0	0	0	6	9	S
5	ME3302	Automation in manufacturing	3	1	0	0	6	10	P
6	ME3482	Mechanical engineering lab 2	0	0	0	3	0	3	P
7	ME3484	Mechanical engineering lab 3	0	0	0	3	0	3	P
		Credits for semester 6						**52**	
		Honors elective 1	3	0	0	0	6	9	
		Credits for semester 6 for honors						**61**	

Branch Code: ME22

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Mechanical Design

2017-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME	Professional Elective 2	3	0	0	0	6	9	P
2	ME5205	Theory of Vibration	3	0	0	0	6	9	P
3	ME5207	Design with Advanced Materials	3	0	0	0	6	9	P
4		Free Elective 4	3	0	0	0	6	9	
5		Free Elective 5	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	2	0	0	0	0	0	H
8	ME5281	Mechanical Design Laboratory	0	0	0	3	0	3	P
		Credits for semester 7						**57**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**66**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME5203	Advanced Mechanics of Solids	3	0	0	0	6	9	P
2		Professional Elective 3	3	0	0	0	6	9	P
3		Professional Elective 4	3	0	0	0	6	9	P
4		Free Elective 6	3	0	0	0	6	9	
5		Free Elective 7	3	0	0	0	6	9	
6	ME5280	Design Practice with CAD Tools	1	0	0	3	2	6	P
		Credits for semester 8						**51**	
	ME	Honours Elective 2	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**60**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490#	Project phase 1	0	0	0	0	22	22	
		Credits for summer						**22**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490+	Project phase 2	0	0	0	0	23	23	
2	ME	Professional Elective 5	3	0	0	0	6	9	P
3		Free Elective 8	3	0	0	0	6	9	
4	ME5204	Finite Element Analysis	3	0	0	0	6	9	P
		Credits for semester 9						**50**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6490$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	57	51	22	50	40	553

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**168+72+85**	**27**	**84**	**72**	**553**

DD Curriculum Mechanical Design

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	240
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**553**

DD Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	240
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**468 + 85**
Percentage		

Note:

- **Professional Elective 1 - to be chosen from 4000 level;**
- **Professional Elective 2-4 - to be chosen from 5000 + level.**

BTech (honours) + M.Tech: (Total credit requirement: 553 + 27 = **580)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME23

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No.31 & 32)

STREAM: Intelligent Manufacturing
2017-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7240	Modeling and Simulation in Manufacturing	3	0	0	0	6	9	P
2	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9	P
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9	P
4	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9	P
5		Free Elective 4	3	0	0	0	6	9	
6	HS	Humanities Elective-3	3	0	0	0	6	9	H
7		Professional Ethics	0	0	0	0	2	0	H
		Credits for semester 7						**54**	
	ME	**Honours elective 2**	3	0	0	0	6	9	
		Credits for semester 7 for Honours						**63**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Free Elective 5	3	0	0	0	6	9	
2		Free Elective 6	3	0	0	0	6	9	
3	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9	P
4	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9	P
5	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4	P
6	ME	Professional Elective 2	3	0	0	0	6	9	P
		Credits for semester 7						**49**	
		Honours elective 3	3	0	0	0	6	9	
		Credits for semester 8 for Honours						**58**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2	ME	Professional Elective 3	3	0	0	0	6	9	P
3	ME	Professional Elective 4	3	0	0	0	6	9	P
4		Free elective 7	3	0	0	0	6	9	
5		Free elective 8	3	0	0	0	6	9	
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	54	49	21	60	40	557

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	45	172+72+85	27	84	72	557

DD Curriculum Intelligent Manufacturing

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	244
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**557**

DD. - Category wise credit Requirements

Category	ME	
	Electives	Total
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	244
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**472 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 557 + 27 = **584)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

Branch Code: ME24

Dual Degree (B.Tech. & M.Tech.) in Mechanical Engineering

(Curriculum for the first six semester is common – refer Page No. 31 & 32)

STREAM: Thermal Engineering
2017-Batch

Semester 7

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME6080	Measurement in Thermal Engg	2	0	0	3	5	10	P
2	ME6150	Numerical Methods in Thermal Engg	2	0	0	3	5	10	P
3		Professional elective 2	3	0	0	0	6	9	P
4		Free elective 4	3	0	0	0	6	9	
5		Free elective 5	3	0	0	0	6	9	
6	HS	Humanities elective 3	3	0	0	0	6	9	H
7		Professional ethics	2	0	0	0	2	0	H
8		**Credits for semester 7**						**56**	
		Honors elective 2	3	0	0	0	6	9	
		Credits for semester 7 for honors						**65**	

Semester 8

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1		Professional elective3	3	0	0	0	6	9	P
2		Professional elective 4	3	0	0	0	6	9	P
3		Professional elective 5	3	0	0	0	6	9	P
4		Professional elective 6	3	0	0	0	6	9	P
5		Professional elective 7	3	0	0	0	6	9	P
6		**Credits for semester 8**						**45**	
		Honors elective 3	3	0	0	0	6	9	
		Credits for semester 8 for honors						**54**	

Summer

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980*	Project phase 1	0	0	0	0	42	21	
		Credits for summer						**21**	

Semester 9

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980	Project phase 2	0	0	0	0	24	24	
2		Free elective 6	3	0	0	0	6	9	
3		Free elective 7	3	0	0	0	6	9	
4		Free elective 8	3	0	0	0	6	9	
5		Professional elective 8	3	0	0	0	6	9	P
		Credits for semester 9						**60**	

Semester 10

S.No	Course No	Course Name	L	T	E	P	O	C	Cat
1	ME7980$	Project phase 3	0	0	0	0	40	40	
		Credits for semester 10						**40**	

Semester	I	II	III	IV	V	VI	VII	VIII	summer	IX	X	Total
Credits	49	56+6	52	59	59	52	56	45	21	60	40	555

Category	Engineering (E)	Professional (P)	Humanities (H)	Sciences (S)	Others	Total
Credits	**45**	**170+72+85**	**27**	**84**	**72**	**555**

DD Curriculum Thermal stream

Category	ME
Basic Science (S)	84
Basic Eng. (E)	45
Profession (P)	242
Humanities (H)	27
Unallocated and project credits	72 + 85
Total	**555**

DD. - Category wise credit Requirements

Category	ME	
	Electives	**Total**
Basic Science (S)	10	84
Basic Eng. (E)	0	45
Profession (P)	72	242
Humanities (H)	27	27
Unallocated and project credits	72 + 85	72 + 85
Total	**181 + 85**	**470 + 85**
Percentage		

BTech (honours) + M.Tech: (Total credit requirement: 555 + 27 = **582)**

- ***Eligibility***: minimum CGPA of 8.5 at the end of 5th sem without U or W grade in any course. They need to maintain these conditions until graduation.
- ***Extra credit requirement***: Additional 27 credits to be taken in VI-VIII from Mechanical department (or equivalent);

M.Tech Curriculum

M.Tech 2016 Batch

Branch Code: ME1

M.Tech. in MECHANICAL ENGINEERING

STREAM: Thermal Engineering

2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6010	Advanced Heat and Mass Transfer	3	0	0	0	6	9
2	ME6020	I.C. Engines Combustion and Pollution	3	0	0	0	6	9
3	ME6030	Refrigeration and Cryogenics	3	0	0	0	6	9
4	ME6040	Incompressible Fluid Flow	3	0	0	0	6	9
5	ME6050	Principles of Turbomachinery	3	0	0	0	6	9
6	ME6140	Applied Thermodynamics	3	0	0	0	6	9
		Total Credits :						**54**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6150	Numerical Methods in Thermal Engineering	2	0	0	3	4	9
2	ME6080	Measurements in Thermal Engineering	2	0	0	3	4	9
3	ME6290	Advanced Energy Conversion	3	0	0	0	6	9
4	ME	Professional Elective I	3	0	0	0	6	9
5	ME	Professional Elective II	3	0	0	0	6	9
6	ME6070	Thermal Engg. Lab	0	0	0	3	0	3
		Total Credits :						**48**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20*
		Total Credits						**20**

** Project (ME7990 *) grade assigned in 4th semester*

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	25	25*
2	ME	Professional Elective III/Free elective	3	0	0	0	6	9
3	ME	Professional Elective IV	3	0	0	0	6	9
4		**Total Credits :**						**43**

** Project (ME7990+) grade assigned in 4th semester*

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	54	48	20	43	40	205

* Free Elective can be taken either in 2nd or 3rd semester

* Grade assigned in fourth semester

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5520	Explosion and Safety	3	0	0	0	6	9
2	ME5530	Thermofluid dynamics of climate	3	0	0	0	6	9
3	ME5550	Flow and Thermal Instabilities	3	0	0	0	6	9
4	ME5560	Heat and Mass Transfer in Biological Systems	3	0	0	0	6	9
5	ME5570	Jetflow & Acoustics	3	0	0	0	6	9
6	ME6001	Theory of Fire Propagation	3	0	0	0	6	9
7	ME6002	Turbomachinery Noise and Control	3	0	0	0	6	9
8	ME6060	Fundamentals of Combustion	3	0	0	0	6	9
9	ME6100	Rocket technology	3	0	0	0	6	9
10	ME6110	Combustion technology	3	0	0	0	6	9
11	ME6120	Air-breathing engines	3	0	0	0	6	9
12	ME6130	Transport Phenomena	3	0	0	0	6	9
13	ME6180	Energy and Environment	3	0	0	0	6	9
14	ME6200	Conduction & Radiation	3	0	0	0	6	9
15	ME6220	Heat exchanger design	3	0	0	0	6	9
16	ME6230	Convection and two-phase flow	3	0	0	0	6	9
17	ME6250	Heat Transfer in Energy Systems	3	0	0	0	6	9
18	ME6270	Design of Power Plant Systems	3	0	0	0	6	9
19	ME6280	Design and Optimization of Energy Systems	3	0	0	0	6	9
20	ME6300	Design of hydro-turbines	3	0	0	0	6	9
21	ME6310	Design of pumps	3	0	0	0	6	9
22	ME6320	Pump Application Engg.	3	0	0	0	6	9
23	ME6330	Cavitation	3	0	0	0	6	9
24	ME6340	Vortex Element Methods	3	0	0	0	6	9
25	ME6350	Mechanical Design of Hydroturbomachines	3	0	0	0	6	9
26	ME6360	Standards in Hydroturbomachines	3	0	0	0	6	9
27	ME6370	Experimental Methods in Hydroturbomachines	3	0	0	0	6	9
28	ME6380	Design of Turbomachines	3	0	0	0	6	9
29	ME6390	Micro Hydro Power	3	0	0	0	6	9
30	ME6400	Design of combustion engines	3	0	0	0	6	9
31	ME6410	Two Stroke Engines	3	0	0	0	6	9
32	ME6420	Simulation of IC Engines Processes	3	0	0	0	6	9
33	ME6430	Engine systems and performance	3	0	0	0	6	9
34	ME6440	Alternative Fuels	3	0	0	0	6	9
35	ME6450	Gas Turbines	3	0	0	0	6	9
36	ME6460	CFD and Its Applications to Engine Processes	3	0	0	0	6	9
37	ME6470	Engine Instrumentation & Electronic Management	3	0	0	0	6	9
38	ME6480	Transport Processes in Engines	3	0	0	0	6	9
39	ME6490	Laser Diagnostics in Engines	3	0	0	0	6	9
40	ME6500	Airconditioning and ventilation	3	0	0	0	6	9
41	ME6510	Refrigeration machinery and components	3	0	0	0	6	9
42	ME6520	Sorption refrigeration and heating systems	3	0	0	0	6	9
43	ME6530	HVAC Systems and Applications	3	0	0	0	6	9
44	ME6540	Food Processing, Storage and Transport	3	0	0	0	6	9
45	ME6550	Vacuum Engineering	3	0	0	0	6	9
46	ME6560	Advanced Cryogenic Systems	3	0	0	0	6	9
47	ME6570	Thermal Energy Conservation	3	0	0	0	6	9
48	ME6580	Utilization of Solar Energy	3	0	0	0	6	9
49	ME6590	Renewable Energy Technology	3	0	0	0	6	9
50	ME6600	Aerodynamic Design of Axial Compressors and Turbines	3	0	0	0	6	9

S.No	Course No	Course Name	L	T	E	P	O	C
51	ME6610	Theory of steam and gas turbines	3	0	0	0	6	9
52	ME6620	Theory and design of centrifugal m/c	3	0	0	0	6	9
53	ME6630	Theory of axial compressors	3	0	0	0	6	9
54	ME6640	Advanced Topics in Turbomachinery	3	0	0	0	6	9
55	ME6650	Computational Fluid Dynamics of Turbomachinery	3	0	0	0	6	9
56	ME6660	Fans, Blowers and Compressors	3	0	0	0	6	9
57	ME6670	Gas Turbine Engineering	3	0	0	0	6	9
58	ME6680	Measurement Techniques in Thermal Turbomachines	3	0	0	0	6	9
59	ME6690	Compact Heat Exchangers	3	0	0	0	6	9
60	ME6910	Diagnostic Methods in Combustion Systems	3	0	0	0	6	9
61	ME6920	Performance Analysis of Hydroturbomachines	3	0	0	0	6	9
62	ME6930	Engine Tribology	3	0	0	0	6	9
63	ME6940	Engine Noise and Vibrations	3	0	0	0	6	9
64	ME7270	Microscale Fluid Flow and Machinery	3	0	0	0	6	9
65	ME7770	Theory & Tech. of Fuel Cells	3	0	0	0	6	9
66	ME7790	Heat and Mass Transfer in Porous Media	3	0	0	0	6	9

Branch Code: ME2

M.Tech. in MECHANICAL ENGINEERING

STREAM: DESIGN

2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6000	Computational Methods in Engineering	3	1	0	0	6	10
2	ME6700	Advanced Mechanics of Solids	3	0	0	0	6	9
3	ME6830	Principles of Product Design	3	0	0	0	6	9
4	ME7360	Theory of Vibration	3	0	0	0	6	9
5	ME7390	Design with Advanced Materials	3	0	0	0	6	9
6	ME6790	CAD and Machine Elements Laboratory	0	0	0	3	0	3
		Total Credits :						**49**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6800	Finite Element Analysis	3	0	0	0	6	9
2	ME7700	Design Practice	1	0	0	3	2	6
3	ME6730	Mechanical Design Lab	0	0	0	3	0	3
4		Professional Elective I	3	0	0	0	6	9
5		Professional Elective II	3	0	0	0	6	9
6		Professional Elective III	3	0	0	0	6	9
7		Professional Elective IV	3	0	0	0	6	9
		Total Credits :						**54**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20*
		Total Credits						**20**

** Project (ME7990 *) grade assigned in 4th semester*

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990+	Project Phase II	0	0	0	0	25	25*
2		Professional Elective V	3	0	0	0	6	9
3		Professional Elective VI	3	0	0	0	6	9
		Total Credits :						**43**

** Project (ME7990+) grade assigned in 4th semester*

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	49	54	20	43	40	206

* Grade assigned in fourth semester

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1	CT7000	Composite Materials Science	3	0	0	0	6	9
2	CT7120	Modeling and Equipment Des for Comp Proc	3	0	0	0	6	9
3	ID5020	Multi-body Dynamics and Applications	3	0	0	0	6	9
4	ID6010	Constitutive Modeling in Continuum Mechanics	3	0	0	0	6	9
5	ID7010	Advanced Finite Element Analysis	3	0	0	0	6	9
6	ME6750	Gear Design	3	0	0	0	6	9
7	ME6760	Design of Mechanical Transmission Elements	3	0	0	0	6	9
8	ME6770	Design of Pressure vessels and Piping	3	0	0	0	6	9
9	ME6780	Design Synthesis	3	0	0	0	6	9
10	ME6810	Transmission Mechanisms and Manipulators	3	0	0	0	6	9
11	ME6840	Design for Manufacture and Assembly	3	0	0	0	6	9
12	ME6850	Product Reliability	3	0	0	0	6	9
13	ME6870	CAD/CAM for product design	3	0	0	0	6	9
14	ME7020	Robotics and Robot applications	3	0	0	0	6	9
15	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9
16	ME7300	Friction and Wear in Machinery	3	0	0	0	6	9
17	ME7400	Mechatronic Systems	3	0	0	0	6	9
18	ME7430	Oil Hydraulics and Pneumatic Systems	3	0	0	0	6	9
19	ME7470	Industrial Instrumentation	3	0	0	0	6	9
20	ME7500	Measurement Systems	3	0	0	0	6	9
21	ME7640	Tribo Design and Analysis	3	0	0	0	6	9
22	ME7660	Nonlinear Solid Mechanics	3	0	0	0	6	9
23	ME7680	Optimization Methods for Mechanical Design	3	0	0	0	6	9
24	ME7710	Advanced Vibration and Acoustics	3	0	0	0	6	9
25	ME7740	Structural Health and Integrity Monitoring	3	0	0	0	6	9
26	ME7820	Rotor Dynamics	3	0	0	0	6	9
27	ME7830	Random Vibrations	3	0	0	0	6	9
28	ME7840	Signal Processing in Mechanical systems	3	0	0	0	6	9
29	ME7850	Modal Analysis of Mechanical systems	3	0	0	0	6	9
30	ME7860	Tribo-Instrumentation	3	0	0	0	6	9
31	ME7870	Diagnostic Maintenance	3	0	0	0	6	9
32	ME7880	Vehicular Vibration	3	0	0	0	6	9
33	ME7890	Advanced Applied Finite Element	3	0	0	0	6	9
34	ME7910	Acoustics and Noise Control	3	0	0	0	6	9
35	ME7920	Applied Finite Element	3	0	0	0	6	9
36	ME7930	Chaotic Vibrations	3	0	0	0	6	9
37	ME 7190	Introduction to Fracture Mechanics	3	0	0	0	6	9
38	ME6012	Mechanics of Human Movement	3	0	0	0	6	9
39	ME6710	Theory of Mechanisms	3	0	0	0	6	9
40	ME6720	Failure Analysis and Design	3	0	0	0	6	9
41	ME6016	Mechanics of Thin Films for Microsystem Design	3	0	0	0	6	9
42	ME6003	Variational Principles in Mechanics	3	0	0	0	6	9
43	ME6820	Fundamentals of Engineering Design	3	0	0	0	6	9
44	ME6015	Elastic waves and ultrasonics	3	0	0	0	6	9
45	ID6070	Mechanics of Viscoelastic materials	3	0	0	0	6	9
46	ME8001	Mechanics of Mixtures	3	0	0	0	6	9
47	ME7023	Foundations of computational materials modelling	3	0	0	0	6	9

Branch Code: ME3

M.Tech. in MECHANICAL ENGINEERING
STREAM: MANUFACTURING ENGINEERING
2016 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7090	Advanced Materials and Processing	3	0	0	0	6	9
2	ME6000	Computational Methods in Engineering	3	1	0	0	6	10
3	ME7010	Microprocessors in Automation	3	0	0	0	6	9
4	ME7040	Computer Aided Design in Manufacturing	3	0	0	0	6	9
5	ME7050	Computer Numerical Control and Adaptive Control	3	0	0	0	6	9
		Total Credits :						**46**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7030	Metrology and Computer Aided Inspection	3	0	0	0	6	9
2	ME7060	Manufacturing and Precision Engineering Lab	0	0	0	4	0	4
3	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9
4	ME7430	Oil Hydraulic and Pneumatic Systems	3	0	0	0	6	9
5		Professional Elective I	3	0	0	0	6	9
6		Professional Elective II	3	0	0	0	6	9
		Total Credits :						**49**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20*
		Total Credits						**20**

** Project (ME7990 *) grade assigned in 4th semester*

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	25	25*
2		Professional Elective III	3	0	0	0	6	9
3		Professional Elective IV	3	0	0	0	6	9
4		**Total Credits :**						**43**

** Project (ME7990+) grade assigned in 4th semester*

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	46	49	20	43	40	198

* Grade assigned in fourth semester

LIST OF ELECTIVES

Electives I and II – EVEN Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7100	Production System Design & Control	3	0	0	0	6	9
2	ME7110	Handling Systems Design & Tooling for Automated Manufacturing and Assembly	3	0	0	0	6	9
3	ME7150	Artificial Intelligence in Manufacturing	3	0	0	0	6	9
4	ME7170	Flexible Manufacturing Systems	3	0	0	0	6	9
5	ME7180	Machine Vision and its Applications	3	0	0	0	6	9
6	HS 7410	Management of Finance, Marketing and Personnel	3	0	0	0	6	9
7	ME7400	Mechatronic Systems	3	0	0	0	6	9
8	ME7410	Control Engineering	3	0	0	0	6	9
9	ME7420	Manufacturing Methods in Precision Engineering	3	0	0	0	6	9
10	ME7470	Industrial Instrumentation	3	0	0	0	6	9
11	ME7480	Optical Instrumentation	3	0	0	0	6	9
12	ME7500	Measurement Systems	3	0	0	0	6	9
13	ME7540	Horological Instruments	3	0	0	0	6	9
14	ME7560	Precision Engineering Elements and Instruments	3	0	0	0	6	9
15	ME6014	Micro Manufacturing Technology	3	0	0	0	6	9
16	ME7020	Robotics and Robot Applications	3	0	0	0	6	9

Electives III and IV – ODD Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	MA6310	Operations Research I	3	0	0	0	6	9
2	ME7160	Computational Methods in Design and Manufacture	3	0	0	0	6	9
3	ME7200	Treatment of Materials	3	0	0	0	6	9
4	ME7210	Machine Tool Dynamics	3	0	0	0	6	9
5	ME7220	Metal Removal Processes	3	0	0	0	6	9
6	ME7300	Friction and Wear in Machinery	3	0	0	0	6	9
7	MT 5610	Metal Forming Processes	3	0	0	0	6	9
8	PH 6470	Applied Optics	3	0	0	0	6	9
9	ME7450	Precision Electric Drives and Actuators	3	0	0	0	6	9
10	ME7460	Inertial Instrumentation	3	0	0	0	6	9
11	ME7490	Design and Packaging of Electronic Equipment	3	0	0	0	6	9
12	ME7510	Precision Engineering Design	3	0	0	0	6	9
13	ME7520	Proportional and Servo Hydraulic Controls	3	0	0	0	6	9
14	ME7530	Design of Robotic Manipulators	3	0	0	0	6	9
15	ME7550	Control System Design: A Computer Aided Approach	3	0	0	0	6	9

M.Tech 2017 Batch

Branch Code: ME1

M.Tech. in MECHANICAL ENGINEERING

STREAM: Thermal Engineering

2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5101	Advanced heat and mass transfer	3	1	0	0	6	10
2	ME5103	Incompressible fluid flow	3	0	0	0	6	9
3	ME5105	Applied Thermodynamics	3	0	0	0	6	9
4	ME5107	Numerical methods in thermal engineering	2	0	0	3	5	10
5	ME5109	Measurements in thermal engineering	2	0	0	3	5	10
		Credits for semester 1						48

Semester 2 (Student must register for at least three out of the five electives during 2nd sem)

S.No	Course No	Course Name	L	T	E	P	O	C
1		Professional elective 1	3	0	0	0	6	9
2		Professional elective 2	3	0	0	0	6	9
3		Professional elective 3	3	0	0	0	6	9
4		Professional elective 4	3	0	0	0	6	9
5		Professional free elective	3	0	0	0	6	9
6	ME5180	Thermal engineering lab	0	0	0	3	0	3
		Credits for semester 2						48

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20
		Total Credits						**20**

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	35	35
		Total Credits :						**35**

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	48	48	20	35	40	191

Ratio of elective-credits to total-credits (excluding project credits): 46.8%

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1.	ME5120	Gas Turbine Engineering	3	0	0	0	6	9
2.	ME5124	Heat and Mass Transfer in Biological Systems	3	0	0	0	6	9
3.	ME5127	Introduction to Atmospheric Science	3	0	0	0	6	9
4.	ME6120	Flow and Thermal Instabilities	3	0	0	0	6	9
5.	ME6122	Jetflow & Acoustics	3	0	0	0	6	9
6.	ME6123	Cavitation	3	0	0	0	6	9
7.	ME6124	Turbomachinery Noise and Control	3	0	0	0	6	9
8.	ME6125	Theory of steam and gas turbines	3	0	0	0	6	9
9.	ME6126	Design of pumps	3	0	0	0	6	9
10.	ME6127	Energy and Environment	3	0	0	0	6	9
11.	ME6128	Aerodynamic Design of Axial Compressors and Turbines	3	0	0	0	6	9
12.	ME6130	Theory and design of centrifugal m/c	3	0	0	0	6	9
13.	ME6132	Theory of axial compressors	3	0	0	0	6	9
14.	ME6134	Airconditioning and ventilation	3	0	0	0	6	9
15.	ME6136	Refrigeration machinery and components	3	0	0	0	6	9
16.	ME6138	Sorption refrigeration and heating systems	3	0	0	0	6	9
17.	ME6139	Simulation of IC Engines Processes	3	0	0	0	6	9
18.	ME6140	HVAC Systems and Applications	3	0	0	0	6	9
19.	ME6141	CFD and Its Applications to Engine Processes	3	0	0	0	6	9
20.	ME6142	Advanced Cryogenic Systems	3	0	0	0	6	9
21.	ME6143	Engine Instrumentation & Electronic Management	3	0	0	0	6	9
22.	ME6144	Thermal Energy Conservation	3	0	0	0	6	9
23.	ME6145	Transport Processes in Engines	3	0	0	0	6	9
24.	ME6146	Utilization of Solar Energy	3	0	0	0	6	9
25.	ME6147	Rocket technology	3	0	0	0	6	9

S.No	Course No	Course Name	L	T	E	P	O	C
26.	ME6148	Renewable Energy Technology	3	0	0	0	6	9
27.	ME6149	Fundamentals of Combustion	3	0	0	0	6	9
28.	ME6154	Design and Optimization of Energy Systems	3	0	0	0	6	9
29.	ME6156	Design of combustion engines	3	0	0	0	6	9
30.	ME6158	Engine systems and performance	3	0	0	0	6	9
31.	ME6160	Alternative Fuels	3	0	0	0	6	9
32.	ME6164	Laser Diagnostics in Engines	3	0	0	0	6	9
33.	ME6166	Combustion technology	3	0	0	0	6	9
34.	ME6168	Air-breathing engines	3	0	0	0	6	9
35.	ME6170	Theory of Fire Propagation	3	0	0	0	6	9
36.	ME7120	Microscale Fluid Flow and Machinery	3	0	0	0	6	9
37.	ME7121	Heat and Mass Transfer in Porous Media	3	0	0	0	6	9
38.	ME8120	Computational Fluid Dynamics of Turbomachinery	3	0	0	0	6	9

Branch Code: ME2

M.Tech. in MECHANICAL ENGINEERING

STREAM: MECHANICAL DESIGN

2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5201	Computational methods in engineering	3	1	0	0	6	10
2	ME5203	Advanced mechanics of solids	3	0	0	0	6	9
3	ME5205	Theory of vibration	3	0	0	0	6	9
4	ME5207	Design with advanced materials	3	0	0	0	6	9
5		Professional elective 1	3	0	0	0	6	9
6	ME5281	Mechanical design lab	0	0	0	3	0	3
		Credits for semester 1						**49**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5204	Finite element analysis	3	0	0	0	6	9
2		Professional elective 2	3	0	0	0	6	9
3		Professional elective 3	3	0	0	0	6	9
4		Professional elective 4	3	0	0	0	6	9
5		Professional free elective	3	0	0	0	6	9
6	ME5280	Design practice using CAD tools	1	0	0	3	2	6
		Credits for semester 2						**51**

Note: (1) Student must register for at least three out of the four electives during the second semester
(2) Student can opt to do professional elective 1 in the third semester

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6490*	Project Phase I	0	0	0	0	20	20
		Total Credits						**20**

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6490	Project Phase II	0	0	0	0	35	35
		Total Credits :						**35**

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6490#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	1	2	Summer	3	4	Total
Credits	49	51	20	35	40	195

Ratio of elective-credits to total-credits (excluding project credits): 45%

LIST OF ELECTIVES

S.No	Course No	Course Name	L	T	E	P	O	C
1	CT7000	Composite Materials Science	3	0	0	0	6	9
2	CT7120	Modeling and Equipment Des for Comp Proc	3	0	0	0	6	9
3	ID5020	Multi-body Dynamics and Applications	3	0	0	0	6	9
4	ID6010	Constitutive Modeling in Continuum Mechanics	3	0	0	0	6	9
5	ID7010	Advanced Finite Element Analysis	3	0	0	0	6	9
6	ME6750	Gear Design	3	0	0	0	6	9
7	ME6760	Design of Mechanical Transmission Elements	3	0	0	0	6	9
8	ME6770	Design of Pressure vessels and Piping	3	0	0	0	6	9
9	ME6780	Design Synthesis	3	0	0	0	6	9
10	ME6810	Transmission Mechanisms and Manipulators	3	0	0	0	6	9
11	ME6840	Design for Manufacture and Assembly	3	0	0	0	6	9
12	ME6850	Product Reliability	3	0	0	0	6	9
13	ME6870	CAD/CAM for product design	3	0	0	0	6	9
14	ME7020	Robotics and Robot applications	3	0	0	0	6	9
15	ME7120	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	6	9
16	ME7300	Friction and Wear in Machinery	3	0	0	0	6	9
17	ME7400	Mechatronic Systems	3	0	0	0	6	9
18	ME7430	Oil Hydraulics and Pneumatic Systems	3	0	0	0	6	9
19	ME7470	Industrial Instrumentation	3	0	0	0	6	9
20	ME7500	Measurement Systems	3	0	0	0	6	9
21	ME7640	Tribo Design and Analysis	3	0	0	0	6	9
22	ME7660	Nonlinear Solid Mechanics	3	0	0	0	6	9
23	ME7680	Optimization Methods for Mechanical Design	3	0	0	0	6	9
24	ME7710	Advanced Vibration and Acoustics	3	0	0	0	6	9
25	ME7740	Structural Health and Integrity Monitoring	3	0	0	0	6	9
26	ME7820	Rotor Dynamics	3	0	0	0	6	9
27	ME7830	Random Vibrations	3	0	0	0	6	9
28	ME7840	Signal Processing in Mechanical systems	3	0	0	0	6	9
29	ME7850	Modal Analysis of Mechanical systems	3	0	0	0	6	9
30	ME7860	Tribo-Instrumentation	3	0	0	0	6	9
31	ME7870	Diagnostic Maintenance	3	0	0	0	6	9
32	ME7880	Vehicular Vibration	3	0	0	0	6	9
33	ME7890	Advanced Applied Finite Element	3	0	0	0	6	9
34	ME7910	Acoustics and Noise Control	3	0	0	0	6	9
35	ME7920	Applied Finite Element	3	0	0	0	6	9
36	ME7930	Chaotic Vibrations	3	0	0	0	6	9
37	ME 7190	Introduction to Fracture Mechanics	3	0	0	0	6	9
38	ME6012	Mechanics of Human Movement	3	0	0	0	6	9
39	ME6710	Theory of Mechanisms	3	0	0	0	6	9
40	ME6720	Failure Analysis and Design	3	0	0	0	6	9
41	ME6016	Mechanics of Thin Films for Microsystem Design	3	0	0	0	6	9
42	ME6003	Variational Principles in Mechanics	3	0	0	0	6	9
43	ME6820	Fundamentals of Engineering Design	3	0	0	0	6	9
44	ME6015	Elastic waves and ultrasonics	3	0	0	0	6	9
45	ID6070	Mechanics of Viscoelastic materials	3	0	0	0	6	9
46	ME8001	Mechanics of Mixtures	3	0	0	0	6	9
47	ME7023	Foundations of computational materials modelling	3	0	0	0	6	9

Any other elective with the approval of the Faculty Advisor/Guide

Branch Code: ME3

M.Tech. in MECHANICAL ENGINEERING
STREAM: MANUFACTURING ENGINEERING
2017 Batch

Semester 1

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5301	Advanced materials processing and characterization	3	0	0	0	6	9
2	ME5303	Computer aided design in manufacturing	3	0	0	0	6	9
3	ME5305	Computer numerical control and adaptive control	3	0	0	0	6	9
4	ME5201	Computational methods in engineering	3	1	0	0	6	10
5		Professional elective 1	3	0	0	0	6	9
	ME5381	Basic manufacturing lab	0	0	0	4	0	4
		Credits for semester 1						**50**

Semester 2

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME5300	Metrology and computer aided design	3	0	0	0	6	9
2	ME5302	Sensors for intelligent manufacturing and condition monitoring	3	0	0	0	6	9
3		Professional elective 2	3	0	0	0	6	9
4		Professional elective 3	3	0	0	0	6	9
5		Professional elective 4	3	0	0	0	6	9
6	ME5380	Manufacturing and precession engineering lab	0	0	0	4	0	4
		Credits for semester 2						**49**

SUMMER

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990*	Project Phase I	0	0	0	0	20	20
		Total Credits						**20**

Semester 3

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990	Project Phase II	0	0	0	0	35	35
		Total Credits :						**35**

Semester 4

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME7990#	Project Phase III	0	0	0	0	40	40
		Total Credits :						**40**

Semester	I	II	Summer	III	IV	Total
Credits	50	49	20	35	40	194

Ratio of elective-credits to total-credits (excluding project credits): 36.4%

LIST OF ELECTIVES

Electives I and II – EVEN Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	ME6320	Mechatronic Systems	3	0	0	0	6	9
2	ME6320	Micro Manufacturing Technology	3	0	0	0	6	9
3	ME6321	Robotics and Robot Applications	3	0	0	0	6	9
4	ME6323	Production System Design & Control	3	0	0	0	6	9
5	ME6324	Artificial Intelligence in Manufacturing	3	0	0	0	6	9
6	ME6326	Machine Vision and its Applications	3	0	0	0	6	9
7	ME6329	Flexible Manufacturing Systems	3	0	0	0	6	9
8	ME6332	Management of Finance, Marketing and Personnel	3	0	0	0	6	9
9	ME7420	Manufacturing Methods in Precision Engineering	3	0	0	0	6	9

Electives III and IV – ODD Semester

S.No	Course No	Course Name	L	T	E	P	O	C
1	MA6310	Operations Research I	3	0	0	0	6	9
2	ME6331	Treatment of Materials	3	0	0	0	6	9
3	ME6333	Metal Removal Processes	3	0	0	0	6	9
4	MM5610	Metal Forming Processes	3	0	0	0	6	9
5	PH6470	Applied Optics	3	0	0	0	6	9

List of courses offered by the department

The syllabus for each of the courses can be found at the academic section website (https://academic.iitm.ac.in/coursecontent) by providing the course number mentioned in the following list. The description of the symbols for the items in the columns 4-9 is given below.

L: Number of lecture hours per week
T: Number of tutorial hours per week
E: Number of extended tutorial hours per week
P: Number of laboratory hours per week
O: Number of hours expected to be spent out side the class room by the student
C: Total number of credits for the course indicating number of hours to be spent by the student per week

The new course numbers in the format MEABXY means:

- **A**:
  - A $<$ 5: Undergraduate level course
  - 5 $\leq$ A $\leq$ 6: Graduate level course
  - A $>$ 6: Advanced graduate level course
- **B**:
  - B=1: Thermal Stream
  - B=2: Design Stream
  - B=3: Manufacturing Stream
- **XY**: A unique two-digit course number
  - Odd Number: Offered in odd semester
  - Even Number: Offered in even semester

Course No (current)	New No.	Course Title (WITH CORRECTED NEW NAMES)	From 2015 batch					
			L	T	E	P	C	O
ME1100	ME1100	Thermodynamics	3	0	0	0	9	6
ME1120	ME1480	Engineering Drawing	1	0	0	3	7	3
ME2220	ME2201	Kinematics and Dynamics of Machinery	3	1	0	0	10	6
ME2260	ME2260	Materials and Design	3	1	0	0	10	6
ME3170	ME3101	Heat Transfer	3	1	0	0	10	6
ME3260	ME3122	Thermal Power Engineering	3	0	0	0	9	6
ME3310	ME3124	Turbomachines	3	0	0	0	9	6
ME3330	ME3126	Internal Combustion Engines	3	0	0	0	9	6
ME3350	ME3201	Design of Machine Elements	3	1	0	0	10	6
ME3940	ME3220	Dynamic Modeling of Engineering Systems	3	0	0	0	9	6
ME2050	ME3281	Machine Drawing Practice	1	0	0	3	7	3
ME3270	ME3481	Mechanical Engg. Lab I	0	0	0	3	3	
ME3280	ME3482	Mechanical Engg. Lab II	0	0	0	3	3	
ME3300	ME3484	Mechanical Engg. Lab III	0	0	0	3	3	
ME4180	ME4120	Automobile Engineering	3	0	0	0	9	6
ME4250	ME4122	Refrigeration & Airconditioning	3	0	0	0	9	6
ME4002	ME4201	Fundamentals of Tribology	3	0	0	0	9	6
ME4003	ME4203	Applied Mechanics of Materials	3	0	0	0	9	6
ME4420	ME4323	Unconventional Mfg Techniques	3	0	0	0	9	6
ME6010	ME5101	Advanced Heat & Mass Transfer	3	0	0	0	9	6
ME6040	ME5103	Incompressible Fluid Flow	3	0	0	0	9	6
ME6150	ME5107	Numerical Methods in Thermal Engg	2	0	0	3	9	4
ME6080	ME5109	Measurement in Thermal Engg	2	0	0	3	9	4
ME4260	ME5123	Cryogenic Systems	3	0	0	0	9	6
ME6670	ME5120	Gas Turbine Engineering	3	0	0	0	9	6
ME4290	ME5121	Fans, Blowers and Compressors	3	0	0	0	9	6
ME4270	ME5122	Solar Heating and Cooling	3	0	0	0	9	6
ME5560	ME5124	Heat and Mass Transfer in Biological Systems	3	0	0	0	9	6
ME4450	ME5125	Fluid Mechanics in Turbomachines	3	0	0	0	9	6
ME5530	ME5127	Introduction to Atmospheric Science	3	0	0	0	9	6
ME6830	ME5128	Principles of Product Design	3	0	0	0	9	6
ME6050	ME5130	Principles of Turbomachinery	3	0	0	0	9	6
ME6030	ME5132	Refrigeration and Cryogenics	3	0	0	0	9	6
ME5003	ME5134	Process Simulation	3	0	0	0	9	6

Course No (current)	New No.	Course Title (WITH CORRECTED NEW NAMES)	From 2015 batch					
			L	T	E	P	C	O
ME6700	ME5203	Advanced Mechanics of Solids	3	0	0	0	9	6
ME6800	ME5204	Finite Element Analysis	3	0	0	0	9	6
ME7360	ME5205	Theory of Vibration	3	0	0	0	9	6
ME7390	ME5207	Design with Advanced Engineering Materials	3	0	0	0	9	6
ME7700	ME5280	Design Practice using CAD tools	1	0	0	3	6	6
ME7700	ME5280	Design Practice	1	0	0	3	6	6
ME6730	ME5281	Mechanical Design Laboratory	0	0	0	3	3	0
ME7030	ME5300	Metrology and Computer Aided Inspection	3	0	0	0	9	6
ME7090	ME5301	Advanced Materials Processing and Characterization	3	0	0	0	9	6
ME7120	ME5302	Sensors for Intelligent Manufacturing and Condition Monitoring	3	0	0	0	9	6
ME7040	ME5303	CAD in Manufacturing	3	0	0	0	9	6
ME7050	ME5305	CNC and Adaptive Control	3	0	0	0	9	6
ME5001	ME5321	Prognostics and Health Management of Machine Tools	3	0	0	0	9	6
ME7010	ME5323	Microprocessors in Automation	3	0	0	0	9	6
ME7060	ME5380	Manufacturing & Precision Engg Lab	0	0	0	4	4	0
ME5550	ME6120	Flow and Thermal Instabilities	3	0	0	0	9	6
ME6008	ME6121	Microfluidics and Microsystems	3	0	0	0	9	6
ME5570	ME6122	Jet Flow and Acoustics	3	0	0	0	9	6
ME6330	ME6123	Cavitation	3	0	0	0	9	6
ME6002	ME6124	Turbomachinery Noise and Control	3	0	0	0	9	6
ME6610	ME6125	Theory of Steam and Gas Turbines	3	0	0	0	9	6
ME6310	ME6126	Design of Hydroturbines	3	0	0	0	9	6
ME6180	ME6127	Energy & Environment	3	0	0	0	9	6
ME6600	ME6128	Aerodyn. Design of Axial Compressors & Turbines	3	0	0	0	9	6
ME6005	ME6129	Solar Energy For Process Heat & Power Generation	3	0	0	0	9	6
ME6620	ME6130	Theory & Designs of Centrifugal Machines	3	0	0	0	9	6
ME6013	ME6131	Boiling Condensation and Two Phase Flow	3	0	0	0	9	6
ME6630	ME6132	Theory of Axial Compressors	3	0	0	0	9	6
ME6011	ME6133	Convective Heat Transfer	3	0	0	0	9	6
ME6500	ME6134	Airconditioning & Ventilation	3	0	0	0	9	6
ME6009	ME6135	Rediative Heat Transfer	3	0	0	0	9	6
ME6510	ME6136	Refrigeration Machinery & Components	3	0	0	0	9	6
ME6520	ME6138	Sorption,Refrgn & Heating Systems	3	0	0	0	9	6
ME6420	ME6139	Simulation of IC Engine Processes	3	0	0	0	9	6

Course No (current)	New No.	Course Title (WITH CORRECTED NEW NAMES)	From 2015 batch					
			L	T	E	P	C	O
ME6560	ME6142	Advanced Cryogenics Systems	3	0	0	0	9	6
ME6470	ME6143	Engine Instrumentation & Electronic Management	3	0	0	0	9	6
ME6570	ME6144	Thermal Energy Conservation	3	0	0	0	9	6
ME6480	ME6145	Transport Process in IC Engines	3	0	0	0	9	6
ME6580	ME6146	Utilisation of Solar Energy	3	0	0	0	9	6
ME6100	ME6147	Rocket Technology	3	0	0	0	9	6
ME6590	ME6148	Renewable Energy Technology	3	0	0	0	9	6
ME6060	ME6149	Fundamentals of Combustion	3	0	0	0	9	6
ME6290	ME6150	Advanced Energy Conversion	3	0	0	0	9	6
ME6006	ME6151	Computational Heat & Fluid Flow	3	0	0	0	9	6
ME6004	ME6152	Micro & Nanoscale Energy Transport	3	0	0	0	9	6
ME6280	ME6154	Design & Optimisation of Energy Systems	3	0	0	0	9	6
ME6400	ME6156	Design of Combustion Engines	3	0	0	0	9	6
ME6430	ME6158	Engine Systems & Performance	3	0	0	0	9	6
ME6440	ME6160	Alternate Fuels for IC Engines	3	0	0	0	9	6
ME6020	ME6162	IC Engine Combustion and Pollution	3	0	0	0	9	6
ME6110	ME6166	Combustion Technology	3	0	0	0	9	6
ME6001	ME6170	Theory of Fire Propogation	3	0	0	0	9	6
ME6710	ME6221	Theory of Mechanisms	3	0	0	0	9	6
ME6760	ME6222	Design of Mechanical Transmission systems	3	0	0	0	9	6
ME6840	ME6223	Design for Manufacture and Assembly	3	0	0	0	9	6
ME6780	ME6224	Design Synthesis	3	0	0	0	9	6
ME6720	ME6225	Failure Analysis and Design	3	0	0	0	9	6
ME6850	ME6226	Product Reliability	3	0	0	0	9	6
ME6003	ME6227	Variational Principles in Mechanics	3	0	0	0	9	6
ME6870	ME6228	CAD/CAM for Product Design	3	0	0	0	9	6
ME6012	ME6230	Mechanics of Human Movement	3	0	0	0	9	6
ME6016	ME6232	Mechanics of Thin Films for Microsystem Design	3	0	0	0	9	6
ME6015	ME6234	Elastic Waves and Ultrasonics	3	0	0	0	9	6
ME6016	ME6236	Mechanics of Thin Films for Microsystem Design	3	0	0	0	9	6
ME6014	ME6320	Micro Manufacturing Technology	3	0	0	0	9	6
ME7020	ME6321	Robotics and Robot Applications	3	0	0	0	9	6
ME7140	ME6322	Comp. Networking & Std. in CIM	3	0	0	0	9	6
ME7100	ME6323	Production System Design & Control	3	0	0	0	9	6

Course No (current)	New No.	Course Title (WITH CORRECTED NEW NAMES)	From 2015 batch					
			L	T	E	P	C	O
ME7170	ME6329	Flexible Manufacturing Systems	3	0	0	0	9	6
ME7200	ME6331	Treatment of Material	3	0	0	0	9	6
ME7400	ME6332	Mechatronic Systems	3	0	0	0	9	6
ME7430	ME6343	Oil Hyd. & Pneumatic Systems	3	0	0	0	9	6
ME6023	ME6345	Solid Freeform Manufacturing	3	0	0	0	9	6
ME7270	ME7120	Microscale Fluid Flow & Machinery	3	0	0	0	9	6
ME7790	ME7121	Heat and Mass Transport in Porous Media	3	0	0	0	9	6
ME7022	ME7122	Droplet and Spray Dynamics	3	0	0	0	9	6
ME7660	ME7220	Nonlinear Solid Mechanics	3	0	0	0	9	6
ME7640	ME7221	Tribology in Design	3	0	0	0	9	6
ME7820	ME7222	Rotor Dynamics	3	0	0	0	9	6
ME7680	ME7223	Optimization Methods for Mechanical Design	3	0	0	0	9	6
ME7850	ME7224	Modal Analysis of Mechanical Systems	3	0	0	0	9	6
ME7840	ME7225	Signal Processing of Mechanical Systems	3	0	0	0	9	6
ME7021	ME7227	Nonlinear Finite Element Analysis of Solild Continua	3	0	0	0	9	6
ME7910	ME7228	Acoustics and Noise Control	3	0	0	0	9	6
ME7023	ME7229	Foundations of Computational Materials Modelling	3	0	0	0	9	6
ME7920	ME7230	Applied Finite Element Analysis	3	0	0	0	9	6
ME7930	ME7232	Chaotic Vibrations	3	0	0	0	9	6
ME7190	ME7234	Introduction to Fracture Mechanics	3	0	0	0	9	6
ME7970	ME7236	Vehicle Systems Design	3	0	0	0	9	6
ME7710	ME7238	Advanced Vibration and Acoustics	3	0	0	0	9	6
ME7830	ME7242	Random Vibrations	3	0	0	0	9	6
ME7023	ME7244	Foundations of Computational Materials Modelling	3	0	0	0	9	6
ME7880	ME7246	Vehicular Vibration	3	0	0	0	9	6
ME6650	ME8120	Computational Fluid Dynamics of Turbomachinery	3	0	0	0	9	6
ME8001	ME8222	Mechanics of Mixtures	3	0	0	0	9	6
WS1010	WS1301	Workshop-I	0	0	0	3	3	0
WS1020	WS1302	Workshop-II	0	0	0	3	3	0